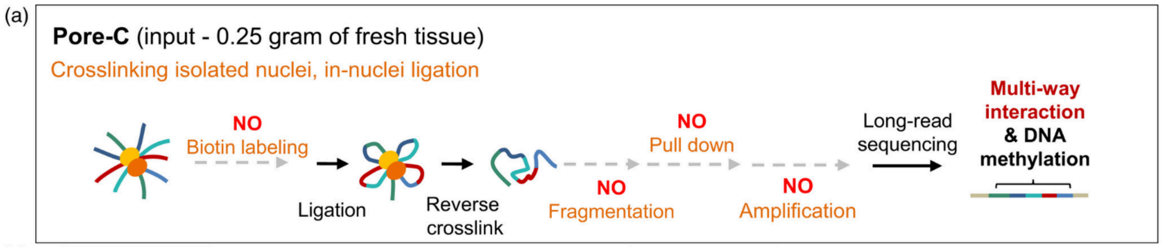
How to choose a suitable enzyme for Pore C?

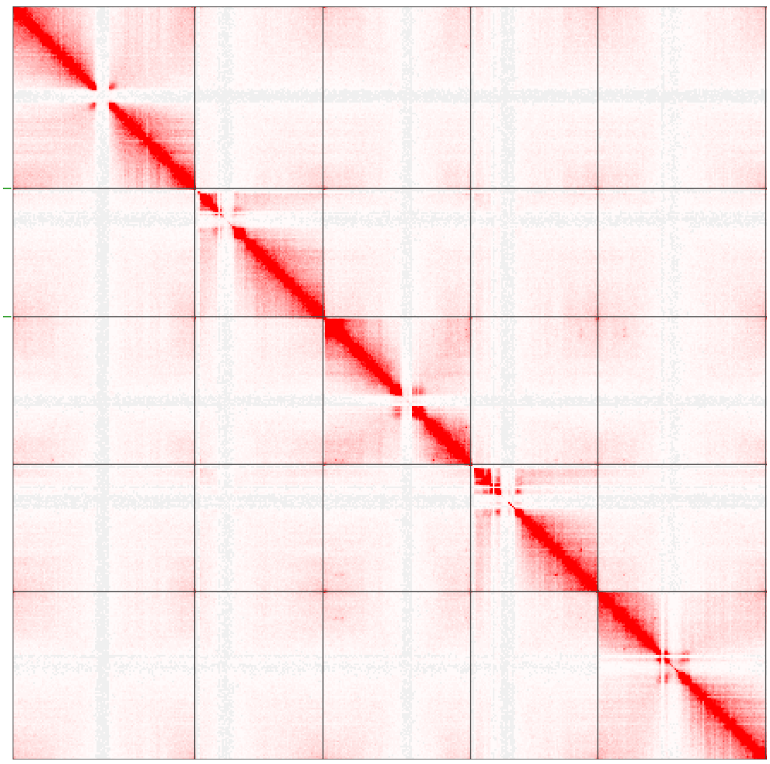
Hanwen, Jacob, Katie and Matt

1. Problems with DpnII

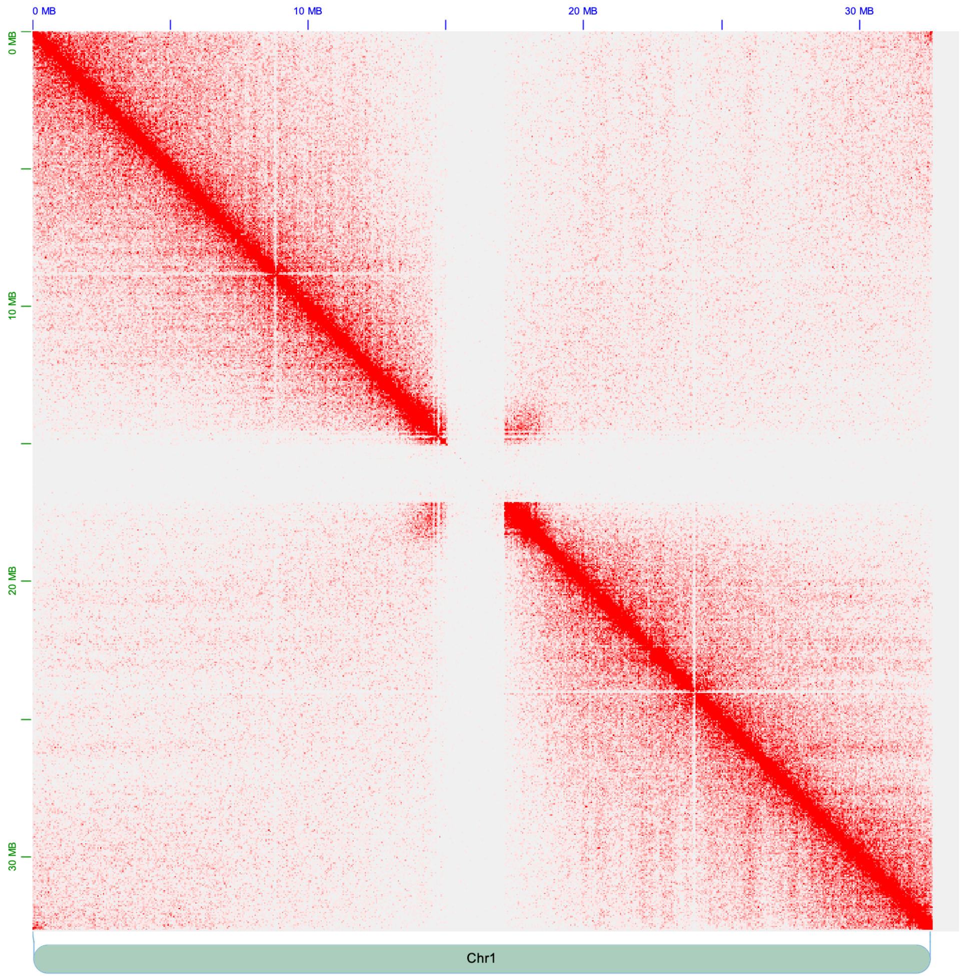
Katie reanalysed PoreC data from
https://onlinelibrary.wiley.com/doi/10.1111/pbi.13811



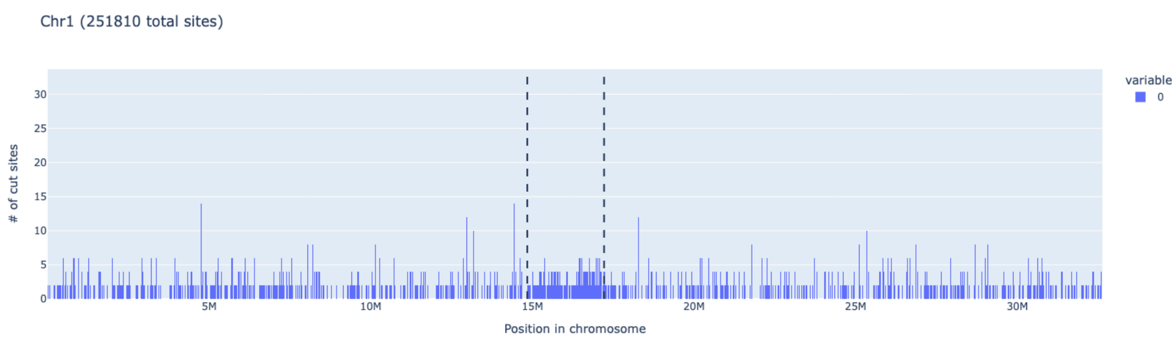
In this paper, the enzyme digestion step was done with DpnI. The signal in the centromeres is almost no present. Here in this first section we show in silico that, as DpnII has a cutting site every CEN178 monomer, the centromeric would be cut evenly but in too short fragments, which would not allow to resolve the 3D contact map with long read sequencing.



- **PoreC map of Chromosome 1 (one of the centromeres with the most depleted signal):**

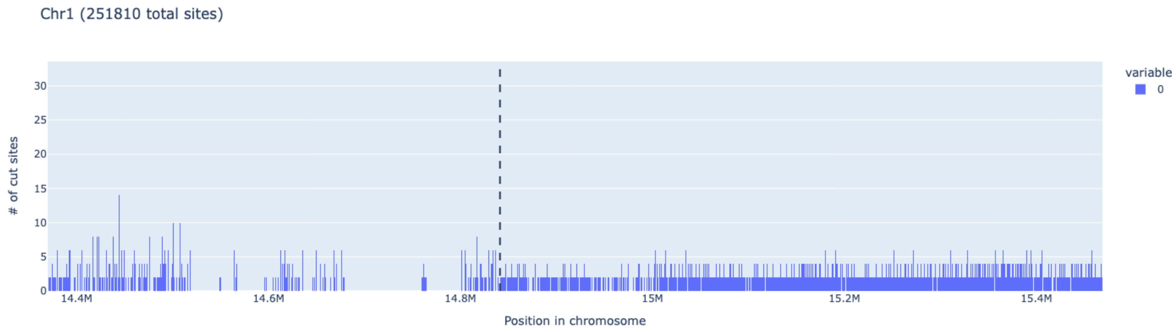


Note below how the centromere 1 is highly cut by DpnII

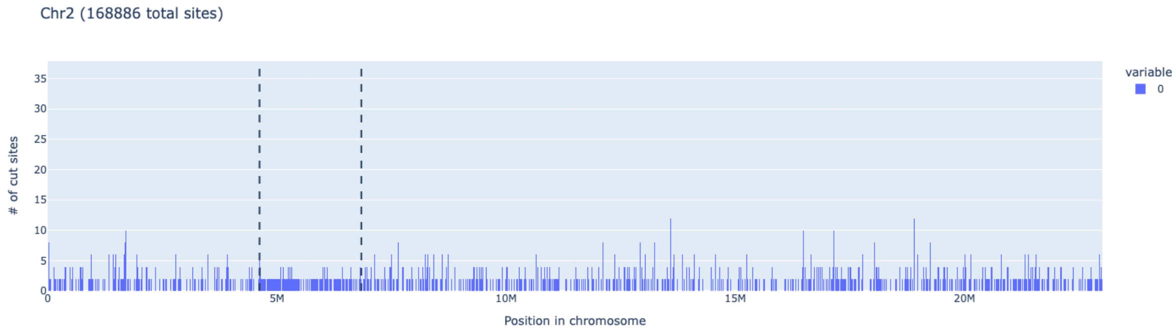


DpnII (GATC) cutting profile in Chromosome 1

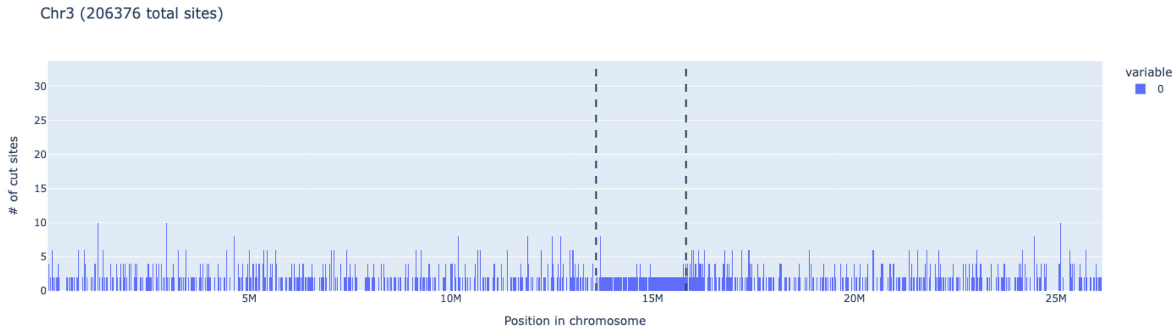
A smaller window size (near 200 bp, that is, more bins) is better to show this phenomenon.



A zoom in of the arm-centromere (p arm) transition to show how different the cutting profile is (Chromosome 1)

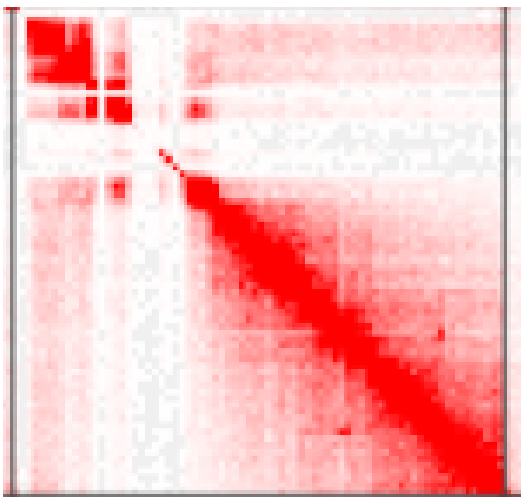


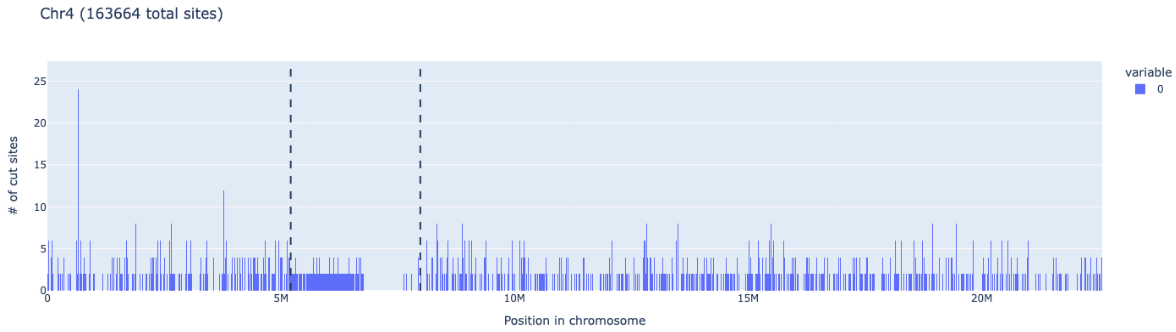
DpnII (GATC) cutting profile in Chromosome 2



DpnII (GATC) cutting profile in Chromosome 3

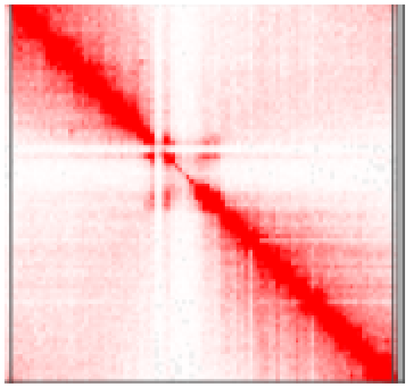
PoreC map of Chromosome 4 (one of the centromere with the least depleted signal): This shows that the least cut part (right part of CEN4), has the best signal.

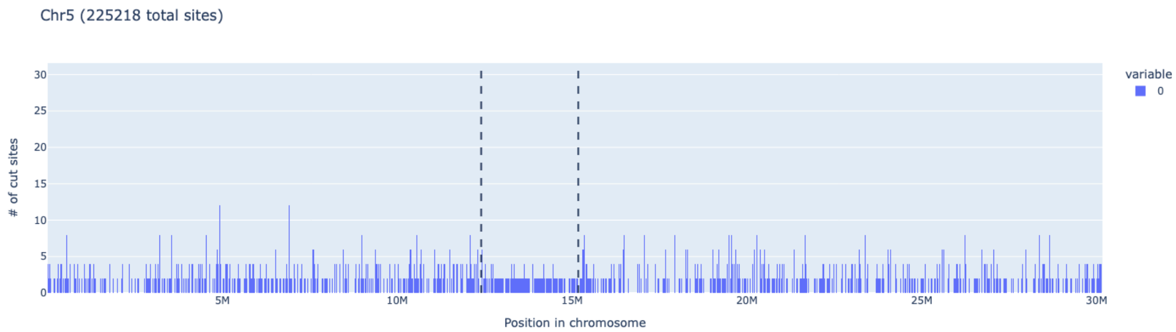




DpnII (GATC) cutting profile in Chromosome 4

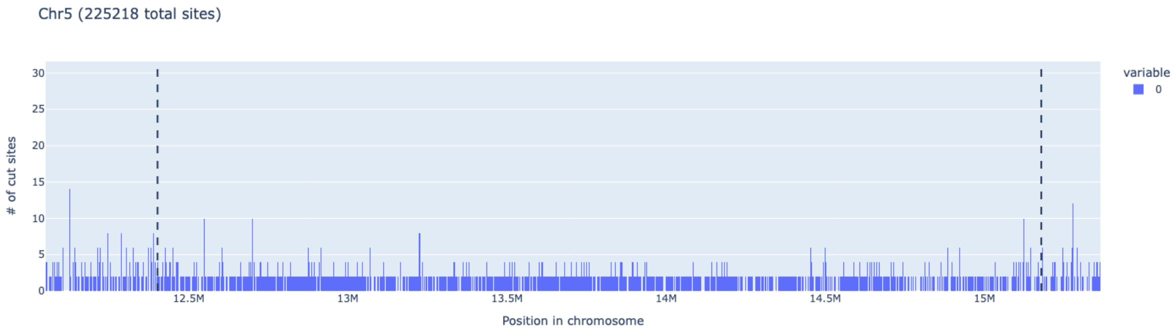
PoreC map of Chromosome 5 (one of the centromere with the least depleted signal):



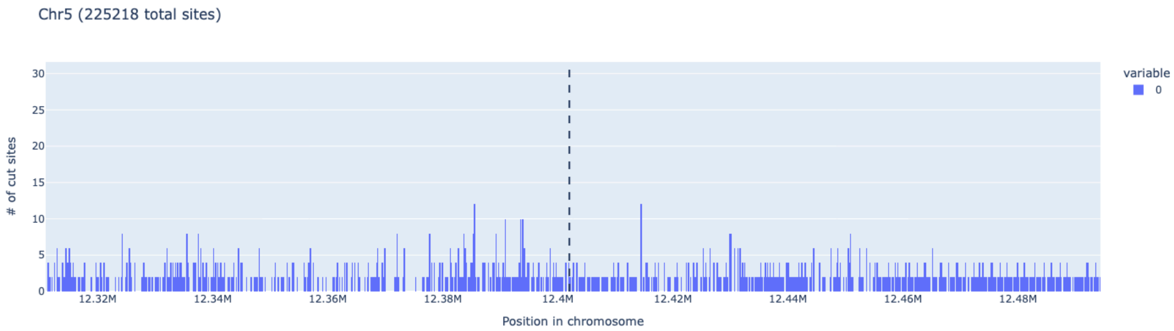


DpnII (GATC) cutting profile in Chromosome 5

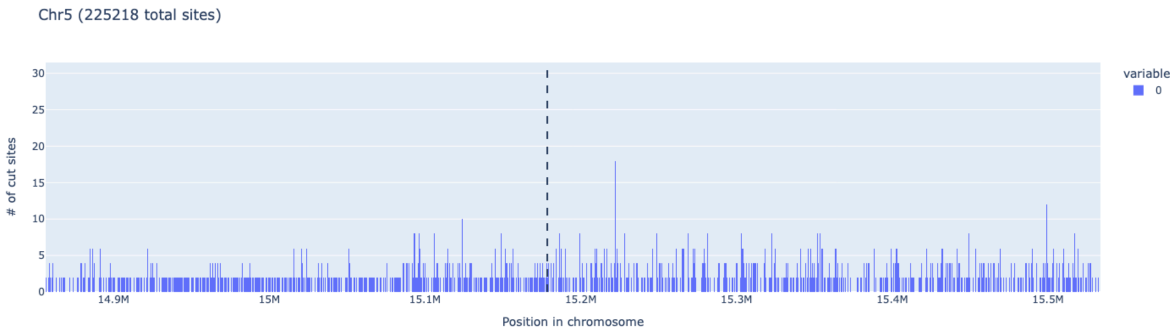
Here a zoom in of CEN5 to show better how it correlates with the PoreC data



A zoom in of the centromere to show the cutting profile is (Chromosome 5)



Arm (p arm)-centromere transition in Chromosome 5



Arm (q arm)-centromere transition in Chromosome 5

2. Our proposed metrics to identify good candidates

2.1 Comparison between number of cuts in ARMS and CEN

After proper normalization (simply taking into account CEN and ARMS lengths), both a ratio or a difference are appropriate, we went forth with difference due to better visualisation on the scatterplot

The **Centromere/Arm Ratio** is a metric that compares the cutting density of a restriction enzyme in the centromeric regions of a chromosome versus its arms. It is calculated as follows:

$$\text{Centromere/Arm Ratio} = \frac{\text{Centromere Density (cuts/kb)}}{\text{Arm Density (cuts/kb)}}$$

where:

- **Centromere Density (cuts/kb)** is the number of cut sites found in the centromeric regions divided by the total length (in kilobases) of those regions.
- **Arm Density (cuts/kb)** is the number of cut sites in the remaining parts of the chromosome (the arms) divided by the length (in kilobases) of those regions.

What Does the Metric Tell Us?

- **Ratio > 1**: A ratio greater than 1 indicates that the enzyme cuts more densely in the centromeric regions compared to the arms. This might suggest that the enzyme's recognition sequence is more common or more accessible in centromeres.
- **Ratio < 1**: A ratio less than 1 suggests that the arms of the chromosome have a higher density of cut sites than the centromeres.
- **Ratio ≈ 1**: A ratio close to 1 means that the cut site density is roughly equal between centromeres and arms.

$$\text{Homogeneity} = \frac{\text{Mean Spacing}}{\text{Std Dev Spacing} + \epsilon}$$

with a small ϵ (here, 1e-6) to avoid division by zero. A higher score indicates more uniform spacing (i.e. lower relative variability).

This is generally true

- **Centromere Homogeneity** is computed as:

$$\text{Cen Homogeneity} = \frac{\text{Centromere Mean Spacing}}{\text{Centromere Spacing Std Dev} + \epsilon}$$

- **Arm Homogeneity** is computed similarly:

$$\text{Arm Homogeneity} = \frac{\text{Arm Mean Spacing}}{\text{Arm Spacing Std Dev} + \epsilon}$$

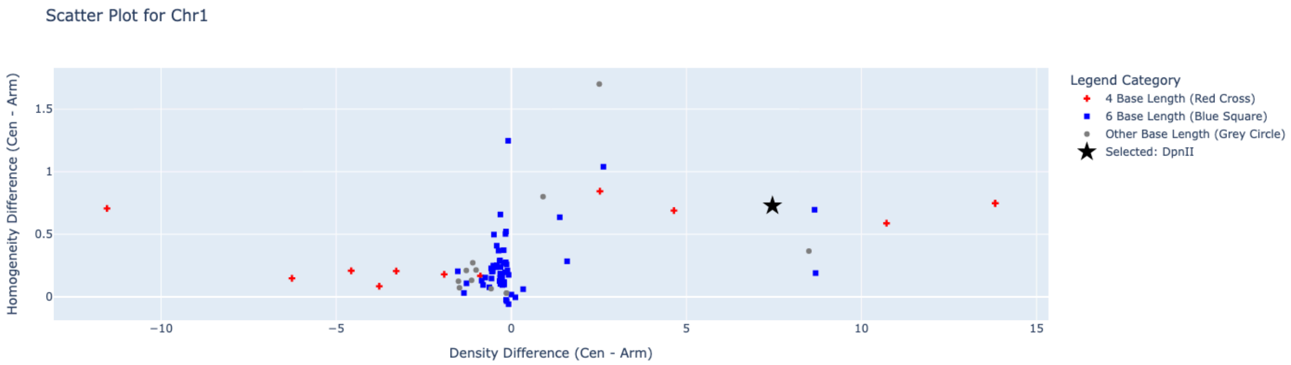
- **Homogeneity Difference (Cen - Arm)** is the subtraction of arm homogeneity from centromere homogeneity.

When it's closer to 0, it should be better (an enzyme whose cutting sites have a similar distribution in ARMS and CEN). However, it is important to not disregard the absolute homogeneity values, because we don't want a centromere mean spacing smaller than 500 bp.

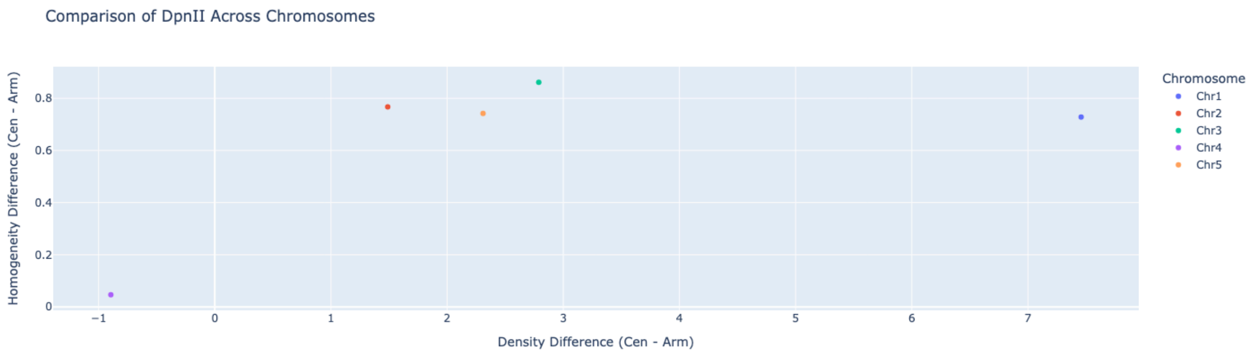
As a reference, the metrics for DpnII per chromosome are:

	A	B	C	D	E	F	G	H	I	J	K	L
1	Chromosome	Enzyme	Total Sites	Centromere Sites	Arm Sites	Centromere Density (cuts/kb)	Arm Density (cuts/kb)	Centromere/Arm Ratio	Centromere Mean Spacing (bp)	Centromere Spacing Std Dev	Arm Mean Spacing (bp)	Arm Spacing Std Dev
2	Chr1	GATC	251810	34756	217054	14.6297071112095	7.17193401175726	2.03985523113102	68.3351172493166	90.1925955849869	150.332144683557	5112.64678912167
3	Chr2	GATC	168886	19282	149604	8.6841108514455	7.1950812353932	1.20695104993779	115.136092526321	145.067787529646	153.788252909367	5815.53504845915
4	Chr3	GATC	206376	23286	183090	10.4432389378626	7.6539765499616	1.3644200331284	95.7471762937513	107.680176171972	142.784962504574	5217.44442855624
5	Chr4	GATC	163664	17928	145736	6.46292075856406	7.35729651157948	0.878436902521492	154.620739666425	2285.22531095712	154.927464233026	7284.13728080263
6	Chr5	GATC	225218	26548	198670	9.56167837205114	7.25218962867891	1.31845399274163	104.581346291483	136.384323636759	151.825403057347	6234.52186720663

Table with detailed metrics for DpnII



DpnII density vs. homogeneity difference by base length in Chr1



Comparison of DpnII density vs. homogeneity across chromosomes

This last plot shows that CEN4 has the best metrics for DpnII (and indeed, its PoreC signal is the "best").

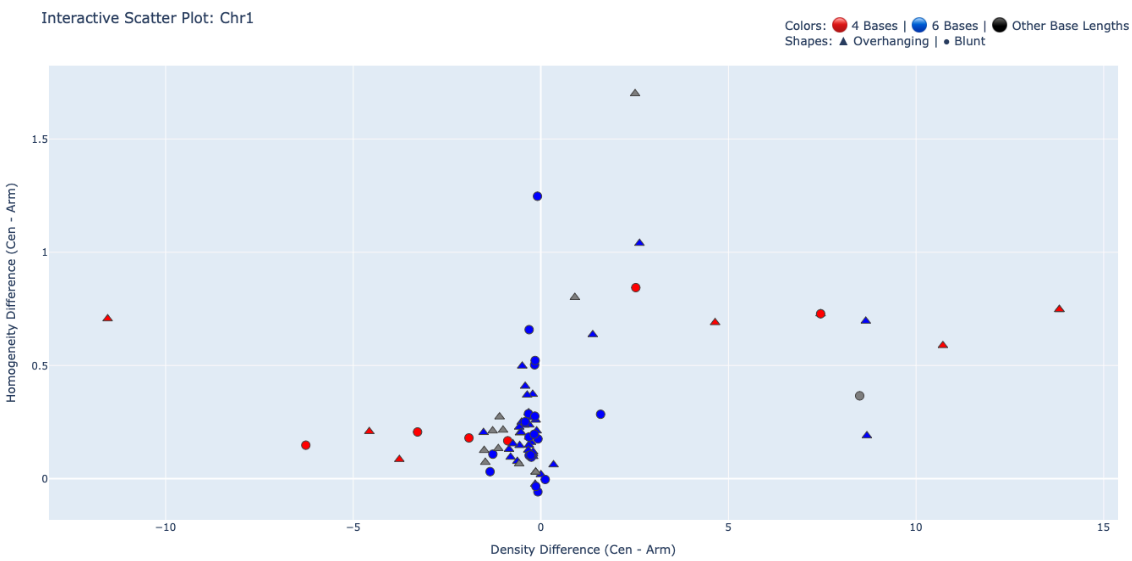
---

3. The proposed enzymes

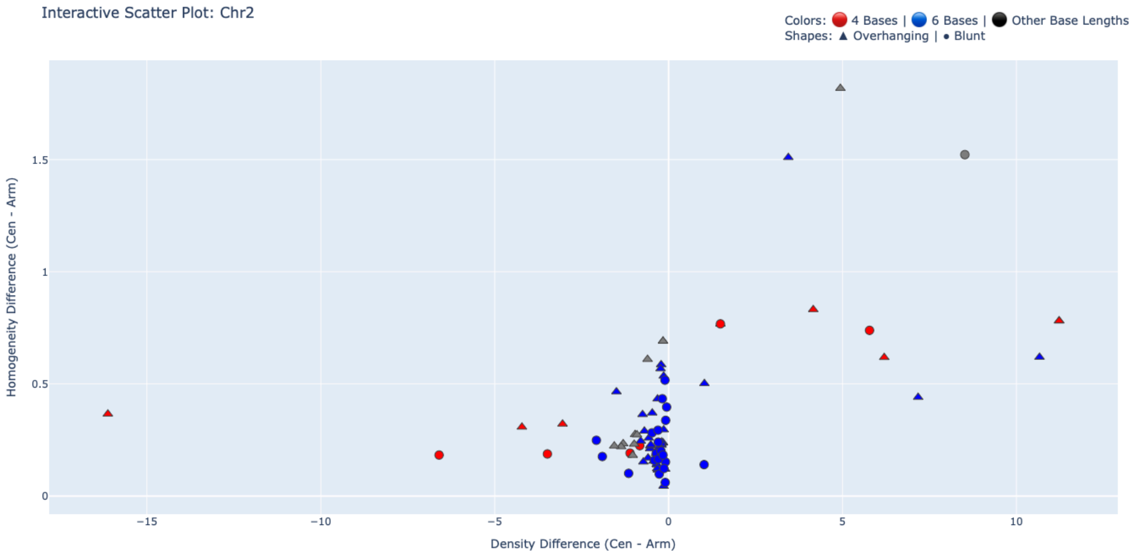
a. Exploration of all NEB enzymes

We downloaded all the set of enzymes from NEB (https://www.neb.com/en-gb/faqs/2016/11/30/how-do-i-create-a-list-of-restriction-enzymes-available-from-neb-in-geneious-software?srsltid=AfmBOooFGZ23BAqdE2JuE2re0dsSKwkVTNxO81K0_oTPBIPxNbvip0YI) and, with Geneious, we filtered out enzymes sensitive to CpG methylation.

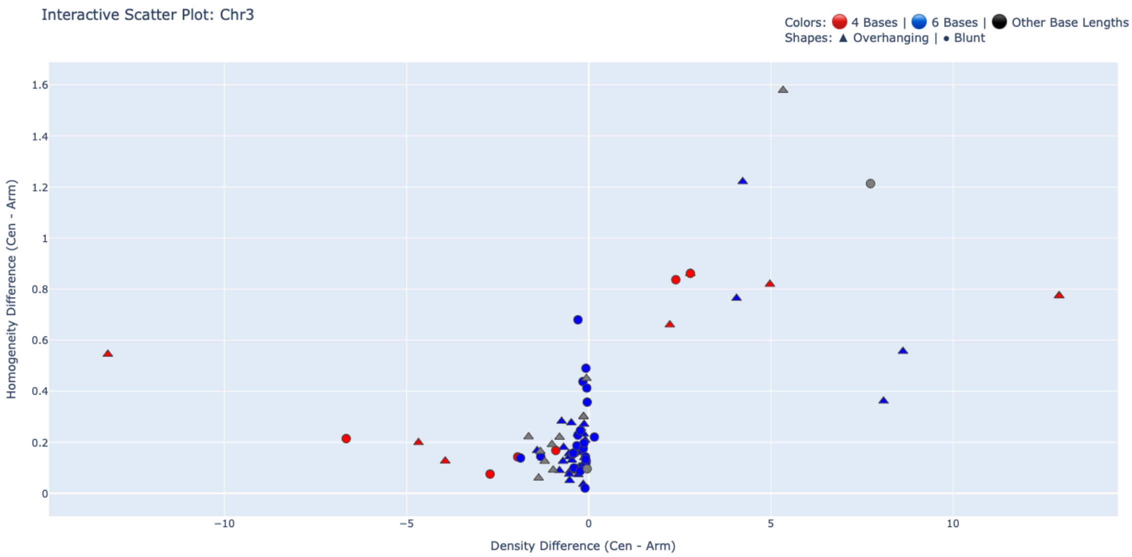
These enzymes are plotted below:



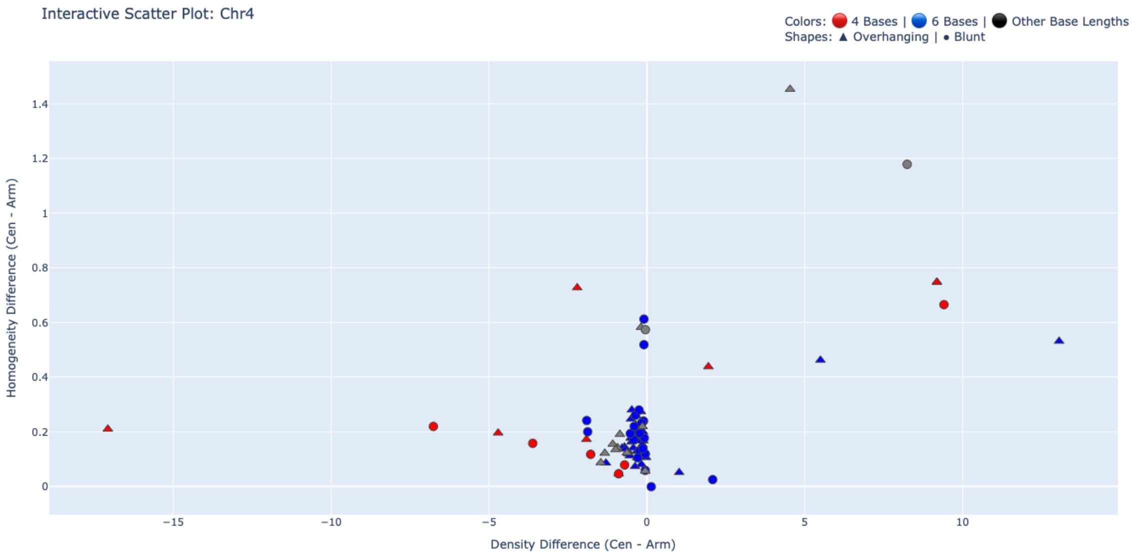
Density vs. Homogeneity Differences in Chr1 by base length



Density vs. Homogeneity Differences in Chr2 by base length



Density vs. Homogeneity Differences in Chr3 by base length



Density vs. Homogeneity Differences in Chr4 by base length

Density vs. Homogeneity Differences in Chr5 by base length

b. Recommended enzymes and their expected performance per chromosome (five 6-cutters, four 4-cutters, one 5 cutter; 7 overhanging, 4 blunt)

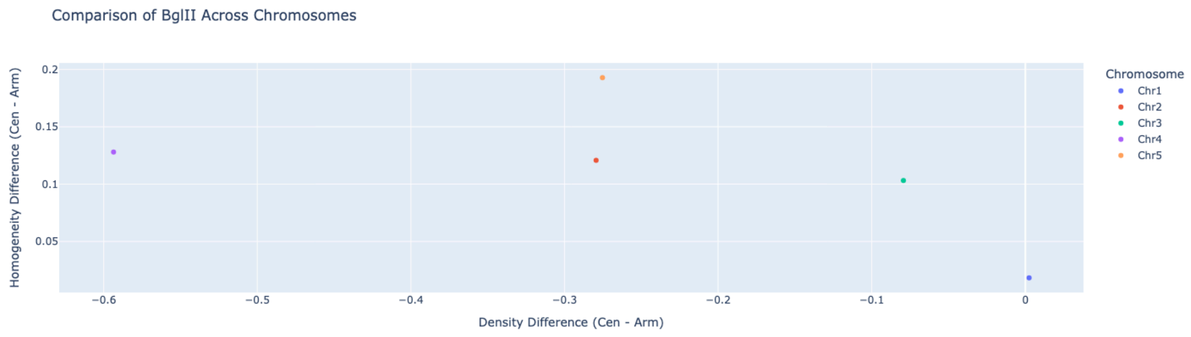
1. BglII (6-cutter; overhanging)
2. AflII (6-cutter; overhanging)
3. BfaI (4-cutter; blunt)
4. BbsI (6-cutter; overhanging)
5. BciVI (6-cutter; overhanging)
6. NdeI (6-cutter; overhanging)

7. AlwI (5-cutter; overhanging)
8. BccI (5-cutter; overhanging)
9. MluCI (4-cutter; blunt)
10. MnlI (4-cutter; blunt)
11. HpyCH4V (4-cutter; blunt)

- **BglII - A^GATCT** (palindrome)

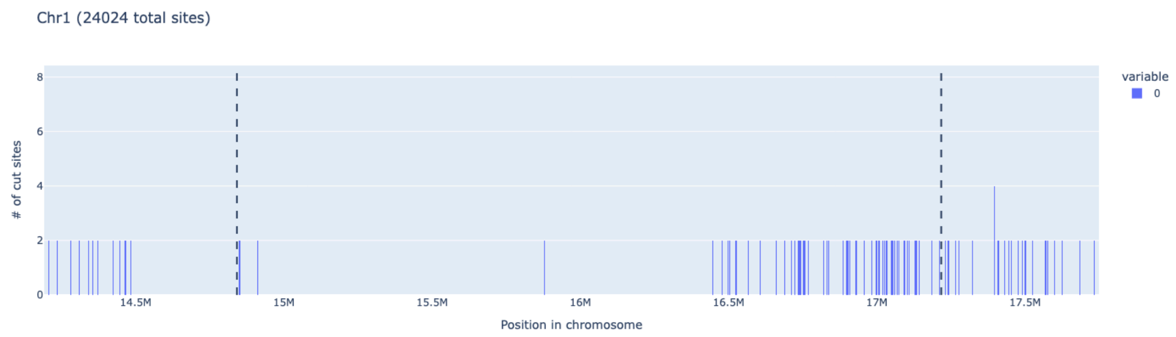
https://www.neb.com/en-gb/products/r0144-bglii?srsltid=AfmBOooYT6yWCD-mOPkqUmWbH3bF5fXwdswbcYEt1qY1QXQFFeLQ4NEF

Nice performance in Chromosome 1 and 3 because they have similar enrichments of cutting sites both in ARMS and CENS. Overall the homogeneity seems adequate (biggest difference is 0.2).

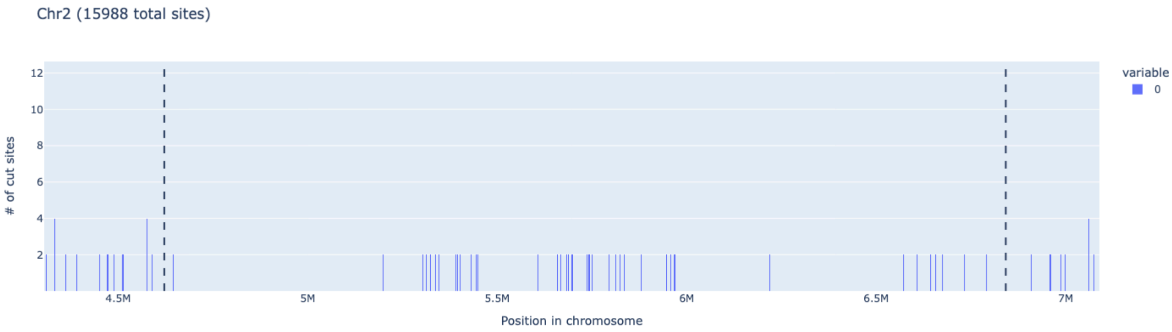


BglII performance across all 5 chromosomes

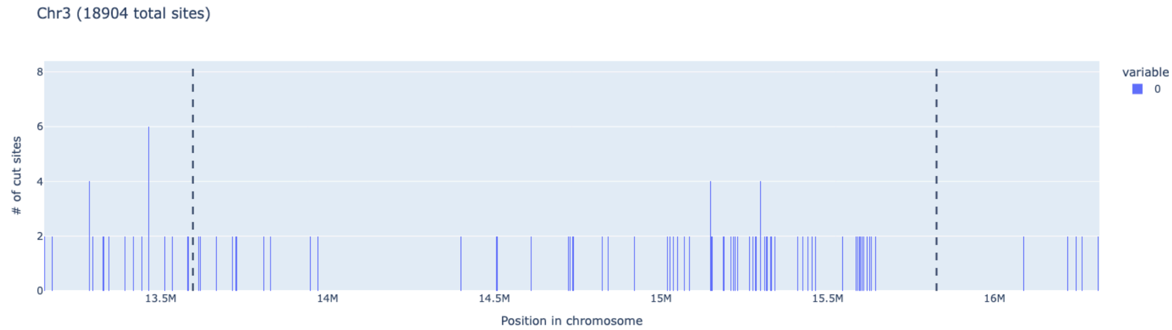
Now, to show how these metrics translate in the chromosome scale view, we show the cutting profile in each chromosome:



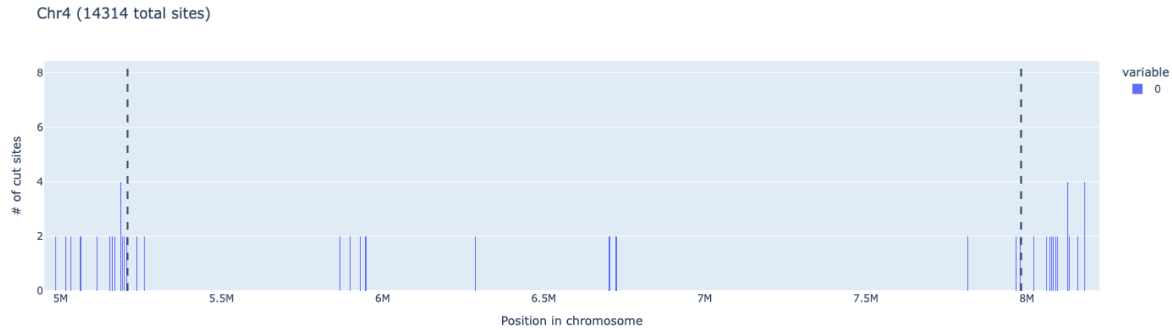
BglII cutting profile in Chromosome 1 (centromere)



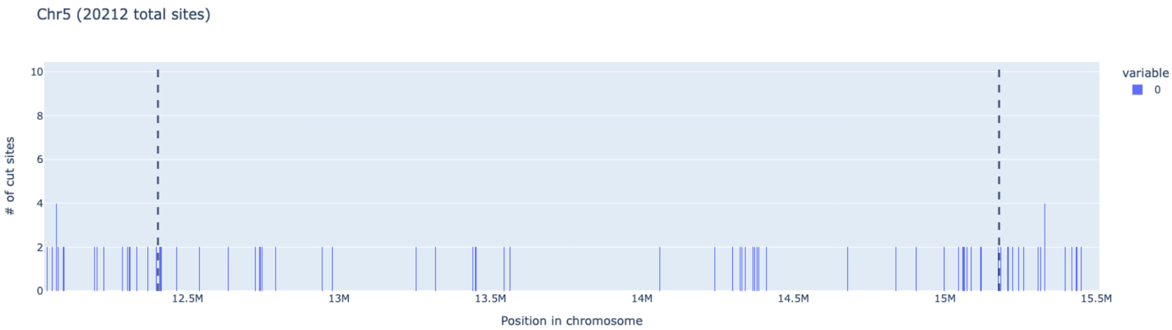
BglII cutting profile in Chromosome 2 (centromere)



BglII cutting profile in Chromosome 3 (centromere)



BglII cutting profile in Chromosome 4 (centromere)

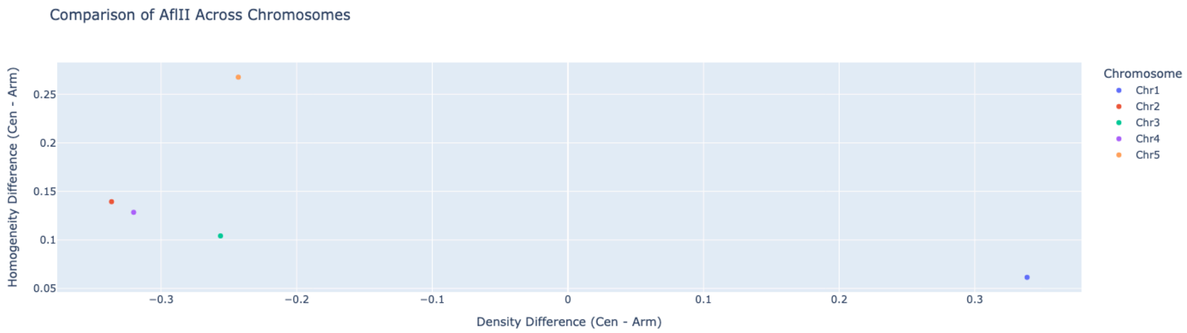


BglII cutting profile in Chromosome 5 (centromere)

Chromosome	Enzyme	Enzyme Recognition	Total Sites	Centromere Sites	Centromere Mean Spacing	Centromere Spacing	Arm Mean Spacing	Arm Spacing Std Dev

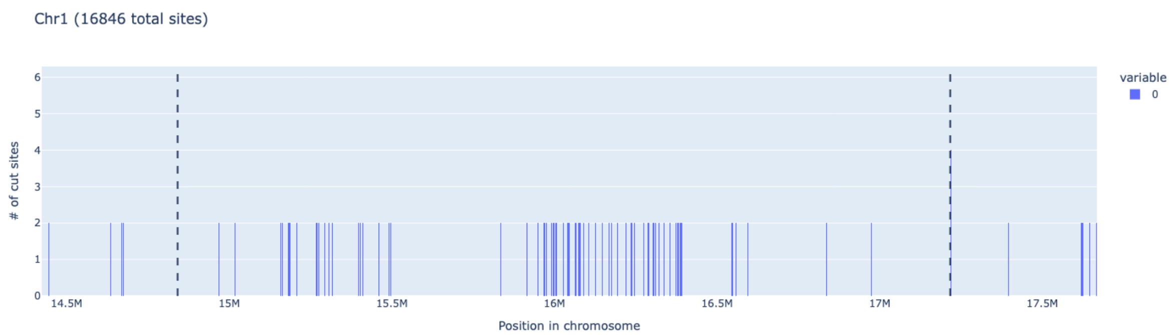
		Sequenc e			(bp)	Std Dev	(bp)	
Chr1	BglII	AGATCT	24024	1754	1354.005 705	12432.16 017	1464.925 053	16157.09 931
Chr2	BglII	AGATCT	15988	982	2212.680 938	10891.60 481	1532.365 012	18591.39 558
Chr3	BglII	AGATCT	18904	1450	1536.786 749	8085.558 125	1497.120 209	17216.00 92
Chr4	BglII	AGATCT	14314	314	8826.194 888	45237.19 332	1612.165 155	24043.52 253
Chr5	BglII	AGATCT	20212	1166	2382.230 901	8865.682 15	1583.761 565	20881.69 983

- **AflII**

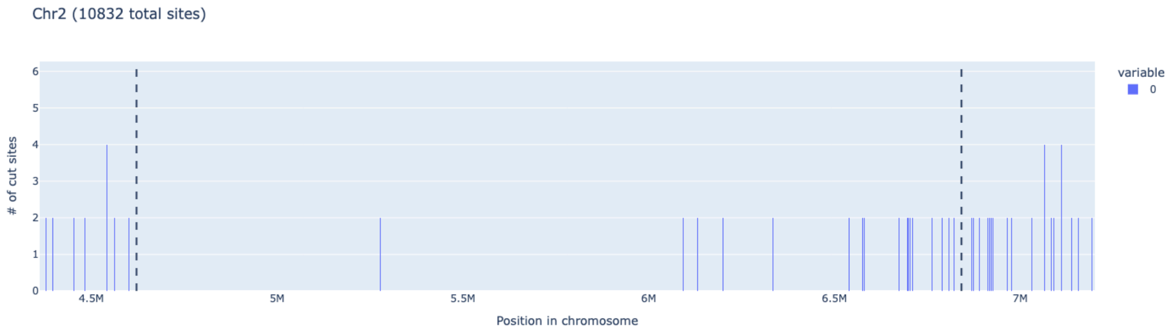


AflII performance across all 5 chromosomes

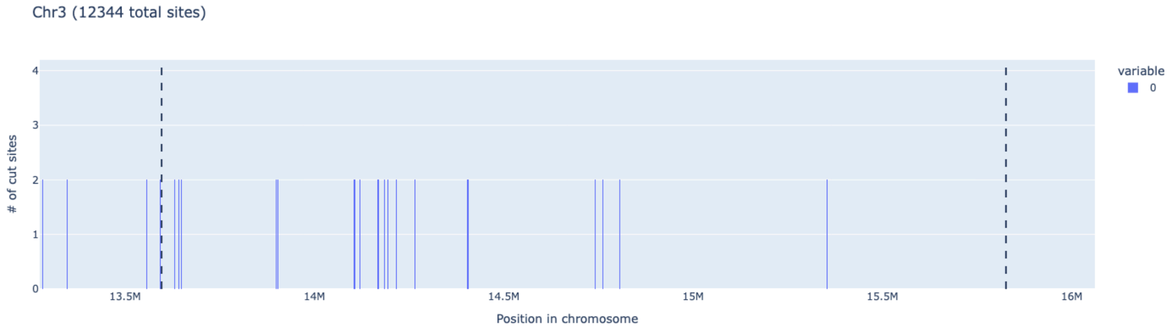
Now, to show how these metrics translate in the chromosome scale view, we show the cutting profile in each chromosome:



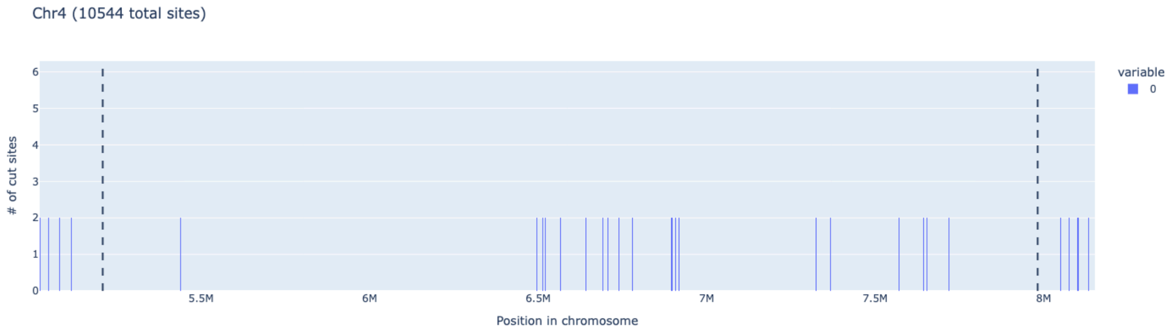
AflII cutting profile in Chromosome 1 (centromere)



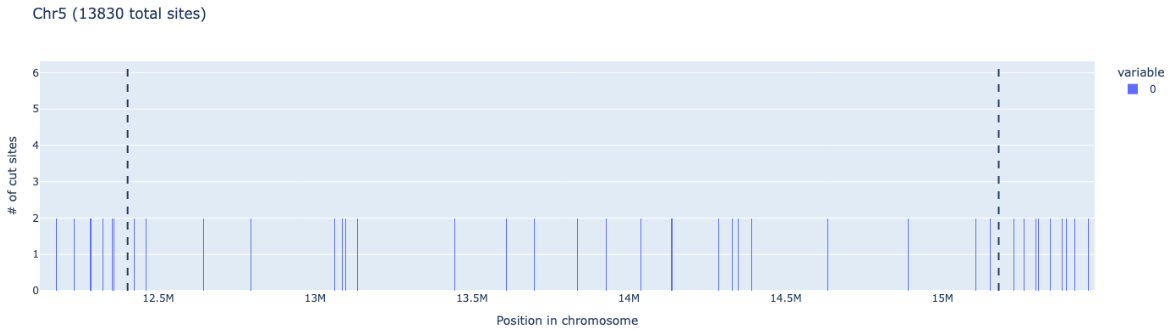
AflII cutting profile in Chromosome 2 (centromere)



AflII cutting profile in Chromosome 3 (centromere)



AflII cutting profile in Chromosome 4 (centromere)

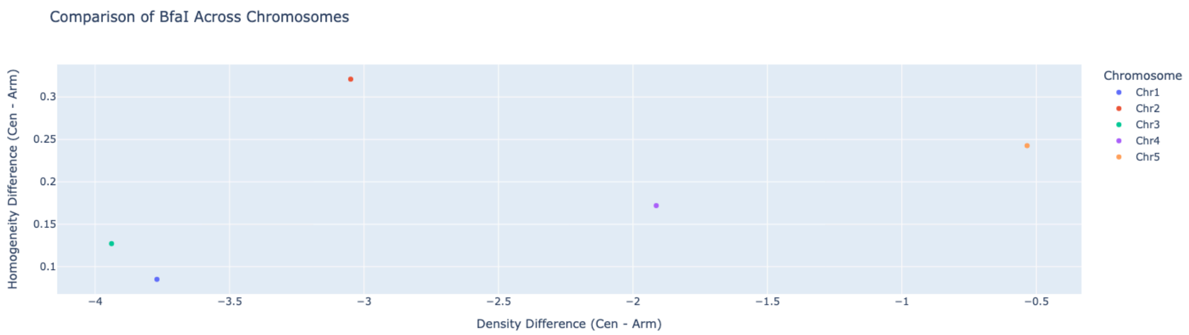


AflII cutting profile in Chromosome 5 (centromere)

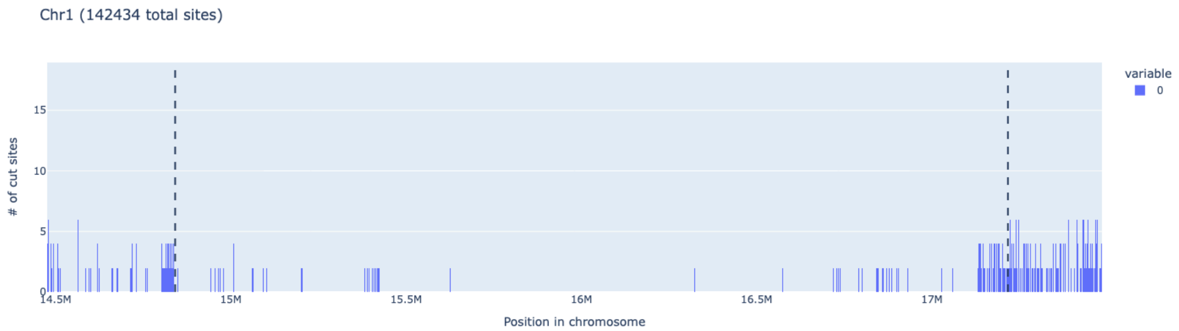
Chromosome	Enzyme	Enzyme Recogniti	Total Sites	Centromere Sites	Centromere Mean	Centromere	Arm Mean	Arm Spacing

		on Sequence			Spacing (bp)	Spacing Std Dev	Spacing (bp)	Std Dev
Chr1	AflII	CTTAAG	16846	1972	1165.075089	6810.691426	2193.87837	20025.26313
Chr2	AflII	CTTAAG	10832	370	5997.655827	25048.56297	2198.451009	21993.21302
Chr3	AflII	CTTAAG	12344	530	4202.508507	20080.12915	2211.721578	21028.18559
Chr4	AflII	CTTAAG	10544	516	5159.145631	24783.38891	2250.612646	28238.6047
Chr5	AflII	CTTAAG	13830	660	4204.383915	11686.45284	2290.284	24874.74775

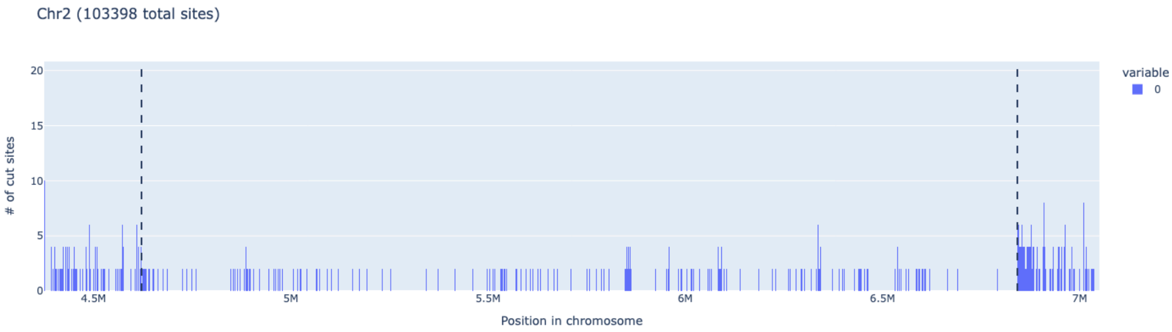
- **BfaI**



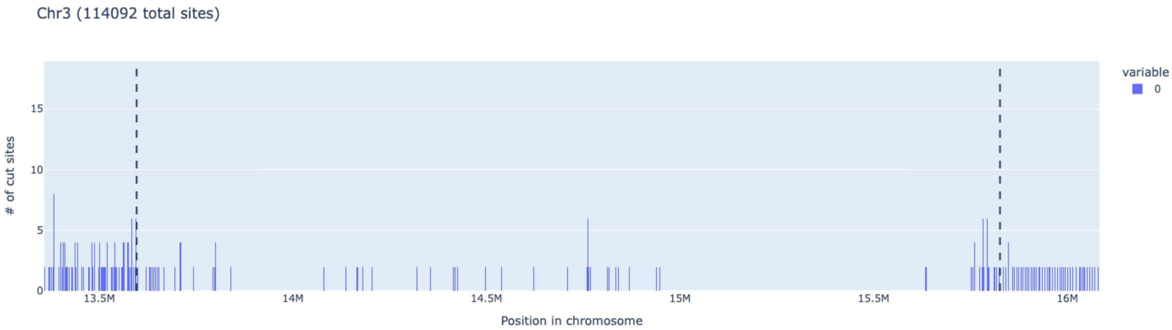
BfaI performance across all 5 chromosomes



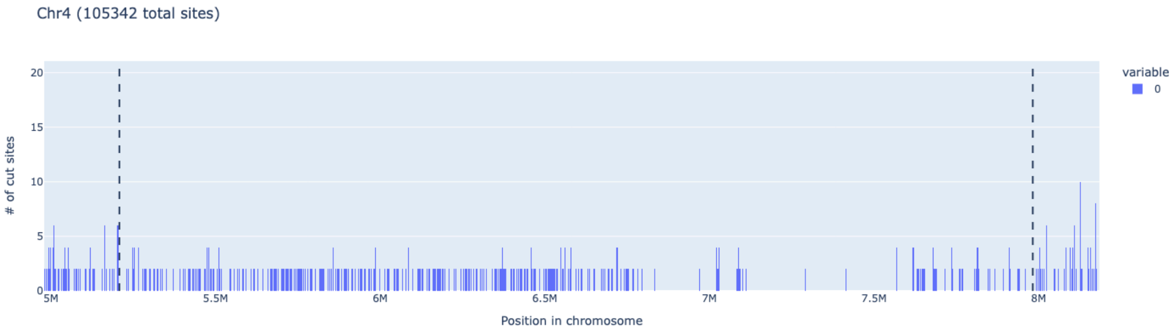
BfaI cutting profile in Chromosome 1 (centromere)



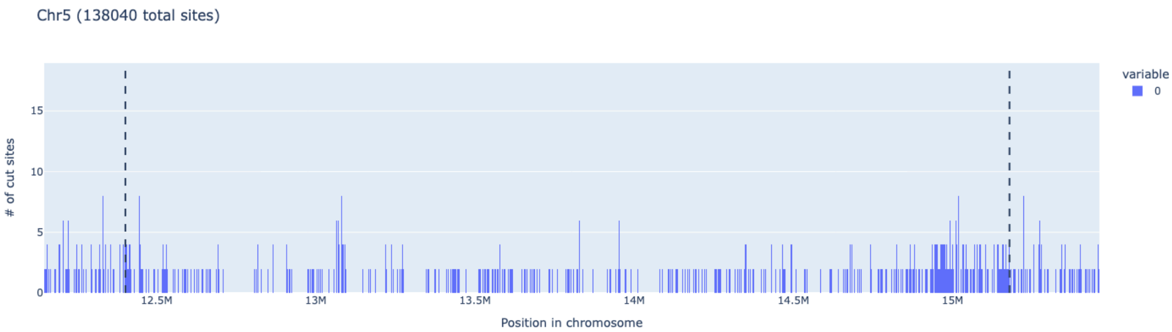
BfaI cutting profile in Chromosome 2 (centromere)



BfaI cutting profile in Chromosome 3 (centromere)



BfaI cutting profile in Chromosome 4 (centromere)

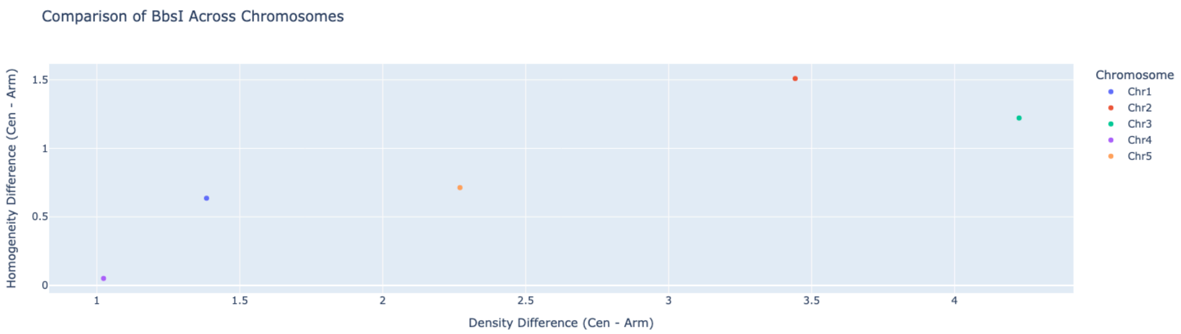


BfaI cutting profile in Chromosome 5 (centromere)

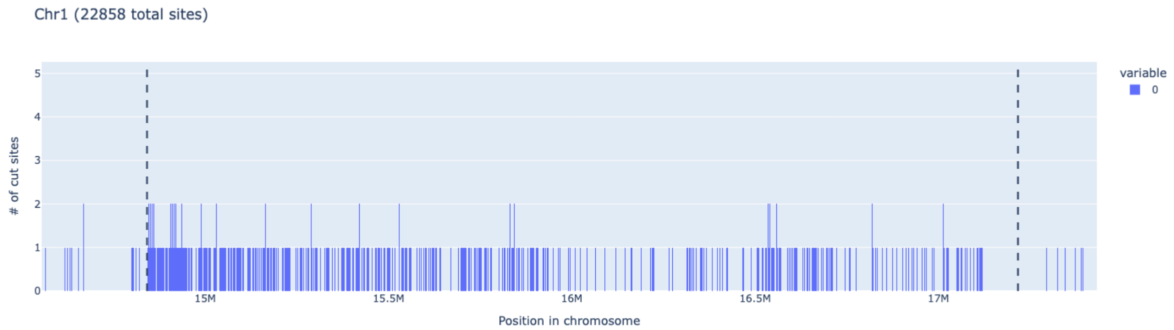
Chromosome	Enzyme	Enzyme Recogniti	Total Sites	Centromere Sites	Centromere Mean	Centromere	Arm Mean	Arm Spacing

		on Sequenc e			Spacing (bp)	Spacing Std Dev	Spacing (bp)	Std Dev
Chr1	BfaI	CTAG	142434	2062	1151.223678	9463.316246	232.4530138	6358.062447
Chr2	BfaI	CTAG	103398	3858	575.0891885	1625.434963	231.1215704	7048.411866
Chr3	BfaI	CTAG	114092	1694	1316.502658	8121.964494	232.58014	6662.109953
Chr4	BfaI	CTAG	105342	8284	332.1300254	1676.148584	232.6254984	8930.947528
Chr5	BfaI	CTAG	138040	11356	244.4863937	894.9681071	238.1013948	7811.31656

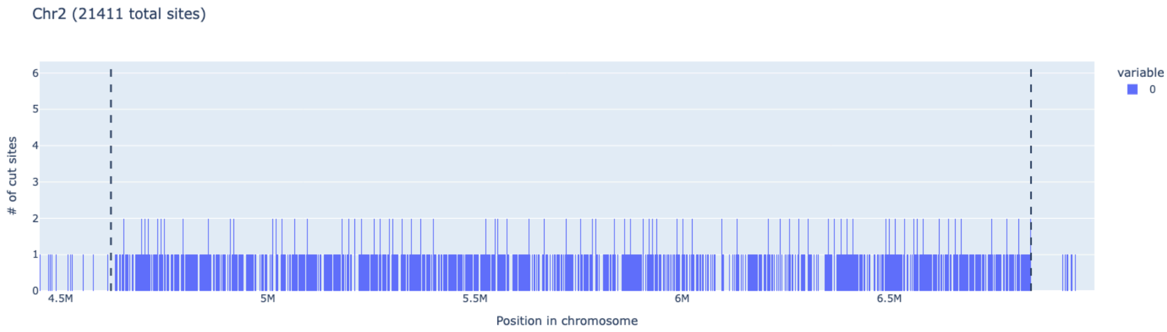
- **BbsI**



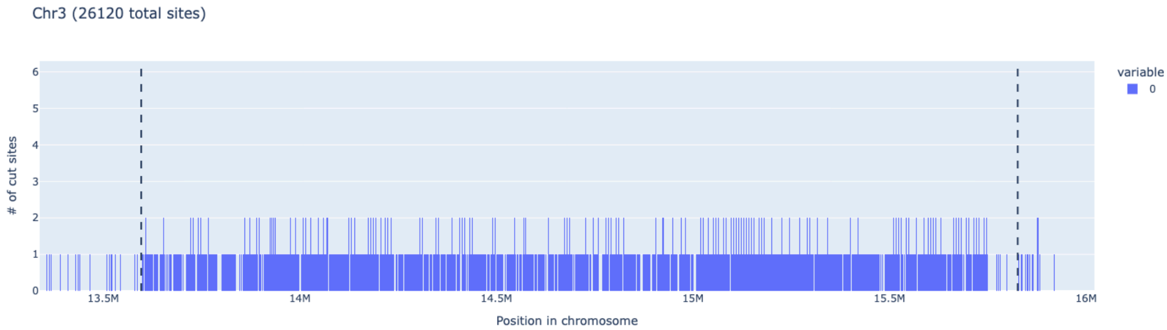
BbsI performance across all 5 chromosomes



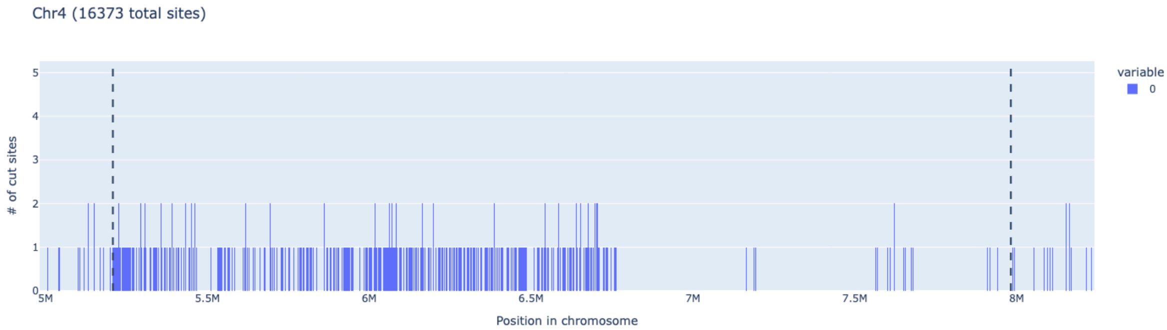
BbsI cutting profile in Chromosome 1 (centromere)



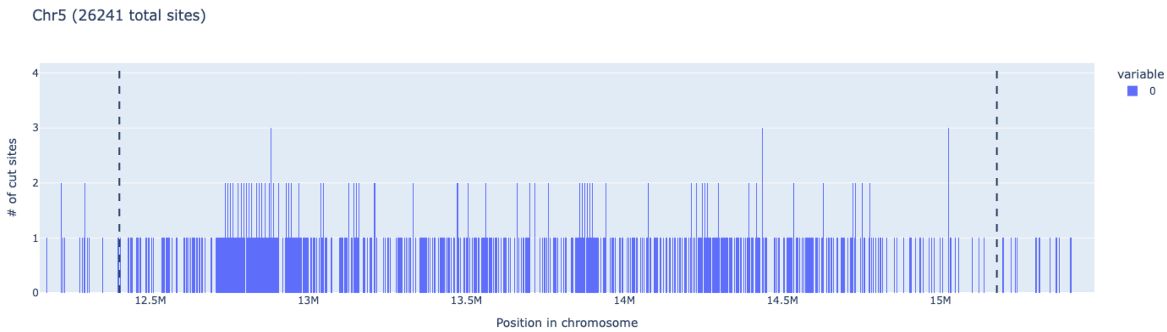
BbsI cutting profile in Chromosome 2 (centromere)



BbsI cutting profile in Chromosome 3 (centromere)



BbsI cutting profile in Chromosome 4 (centromere)

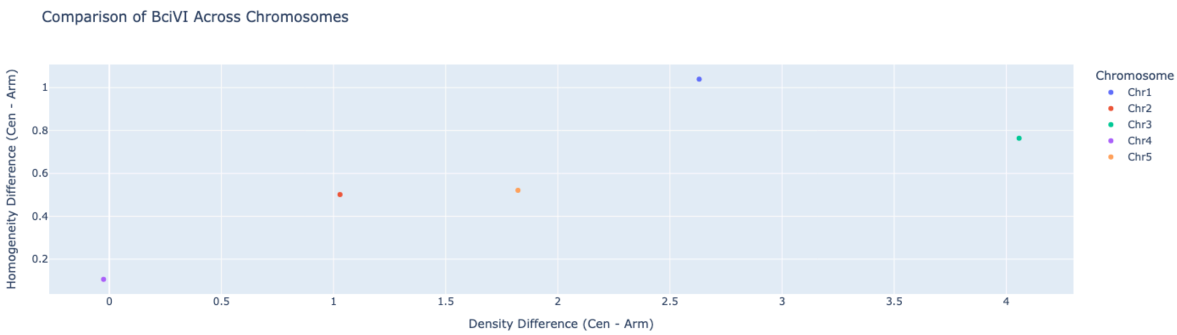


BbsI cutting profile in Chromosome 5 (centromere)

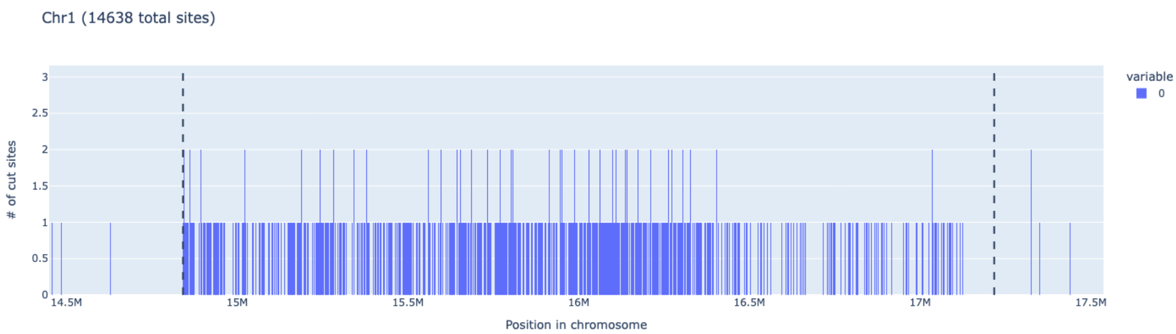
Chromosome	Enzyme	Enzyme Recogniti	Total Sites	Centromere Sites	Centromere Mean	Centromere	Arm Mean	Arm Spacing

		on Sequence			Spacing (bp)	Spacing Std Dev	Spacing (bp)	Std Dev
Chr1	BbsI	GAAGAC	22858	4712	496.9607302	676.4514484	1798.292587	18282.90053
Chr2	BbsI	GAAGAC	21411	8972	247.319028	154.4823247	1849.319022	20197.05262
Chr3	BbsI	GAAGAC	26120	10846	205.2546796	155.9970146	1711.631965	18154.18401
Chr4	BbsI	GAAGAC	16373	4502	607.1746279	4855.612632	1901.657203	25722.82047
Chr5	BbsI	GAAGAC	26241	8138	340.6233255	429.1503898	1666.016683	20711.95323

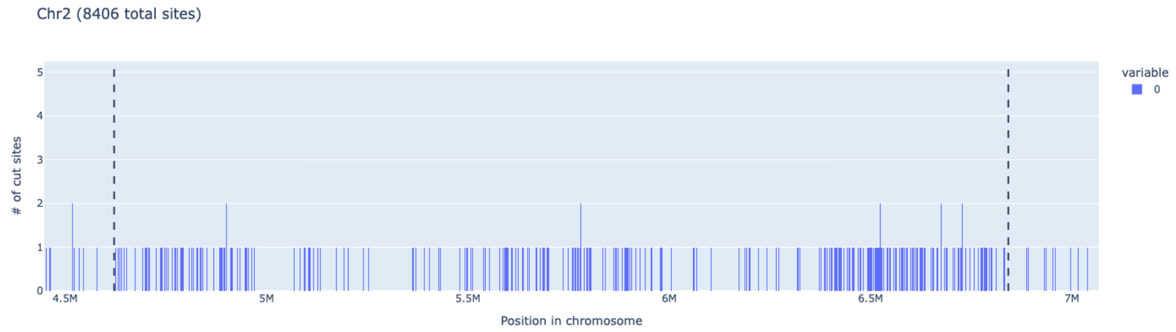
- **BciVI**



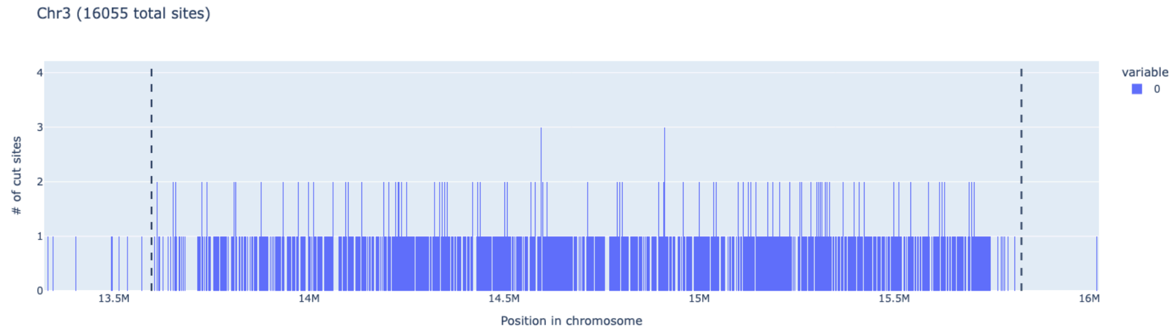
BciVI performance across all 5 chromosomes



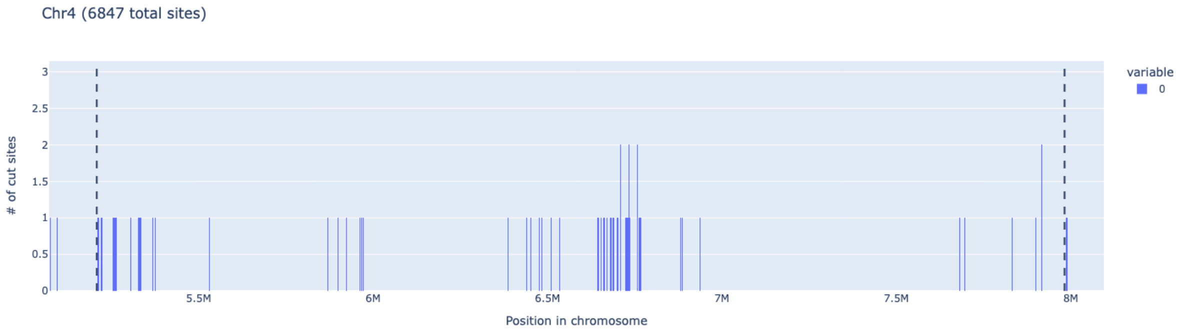
BciVI cutting profile in Chromosome 1 (centromere)



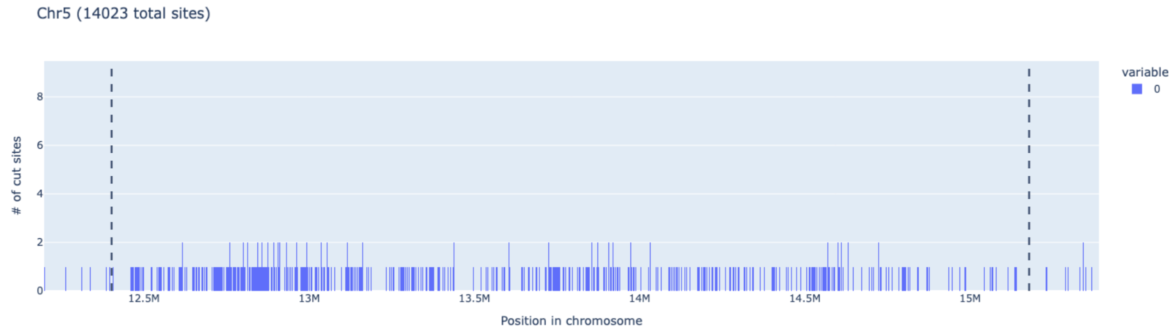
BciVI cutting profile in Chromosome 2 (centromere)



BciVI cutting profile in Chromosome 3 (centromere)



BciVI cutting profile in Chromosome 4 (centromere)

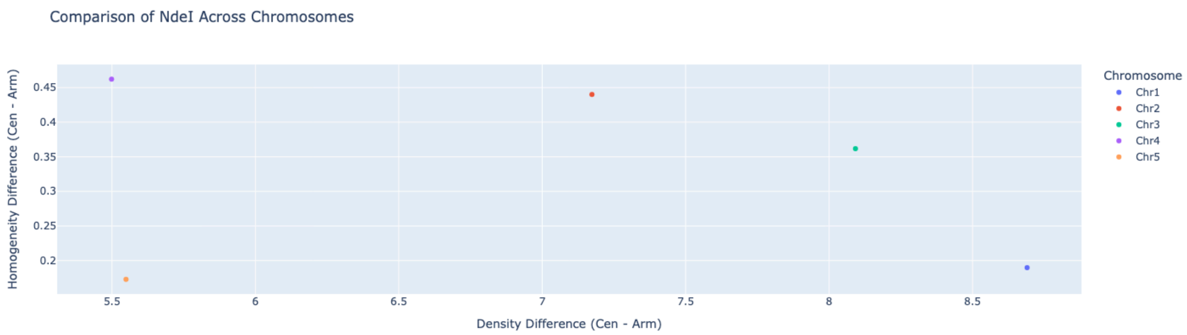


BciVI cutting profile in Chromosome 5 (centromere)

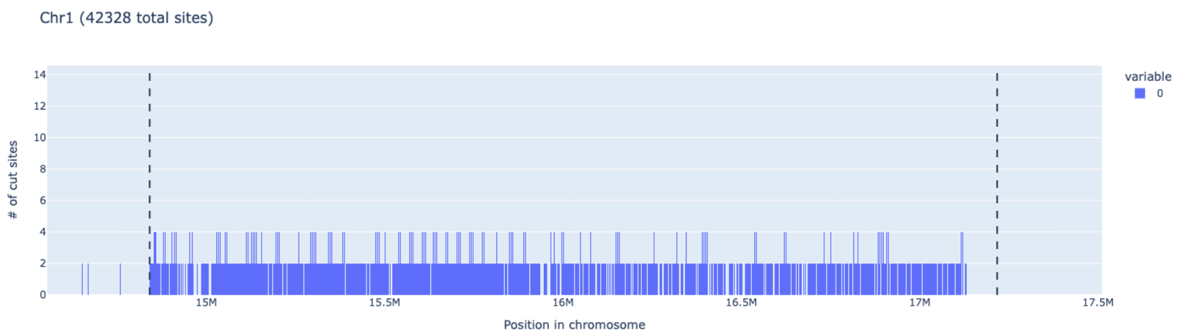
Chromosome	Enzyme	Enzyme Recogniti	Total Sites	Centromere Sites	Centromere Mean	Centromere	Arm Mean	Arm Spacing

		on Sequenc e			Spacing (bp)	Spacing Std Dev	Spacing (bp)	Std Dev
Chr1	BciVI	GTATCC	14638	6860	333.5341887	279.47761	4195.808667	27298.91877
Chr2	BciV	GTATCC	8406	2875	771.5824635	1207.574748	4159.249729	30282.43417
Chr3	BciV	GTATCC	16055	9644	230.6816343	253.9185217	4076.982995	28312.49542
Chr4	BciV	GTATCC	6847	779	3563.724936	16997.88021	3720.986155	35864.25384
Chr5	BciV	GTATCC	14023	5885	471.7367437	736.8510088	3705.598501	31071.98606

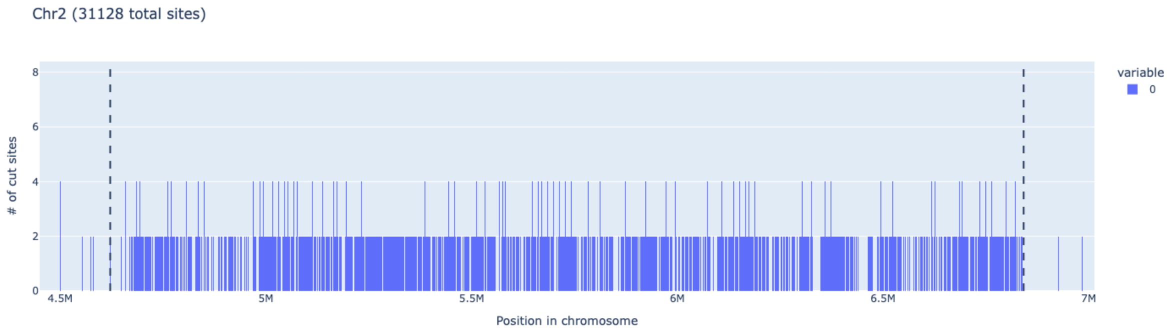
- **NdeI**



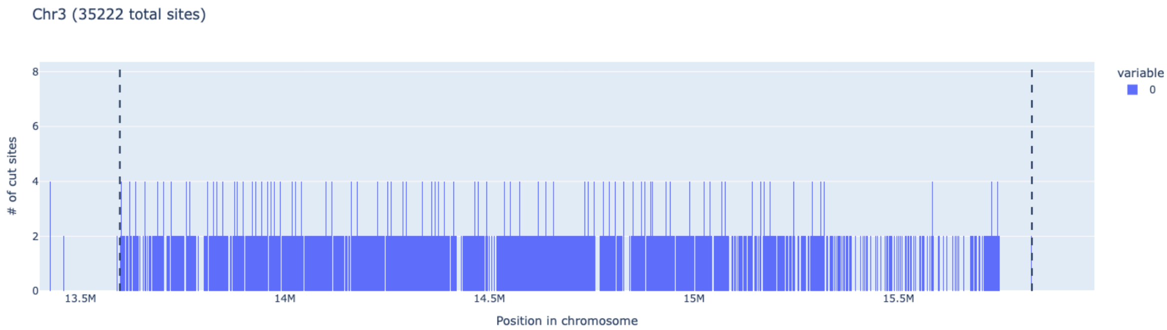
NdeI performance across all 5 chromosomes



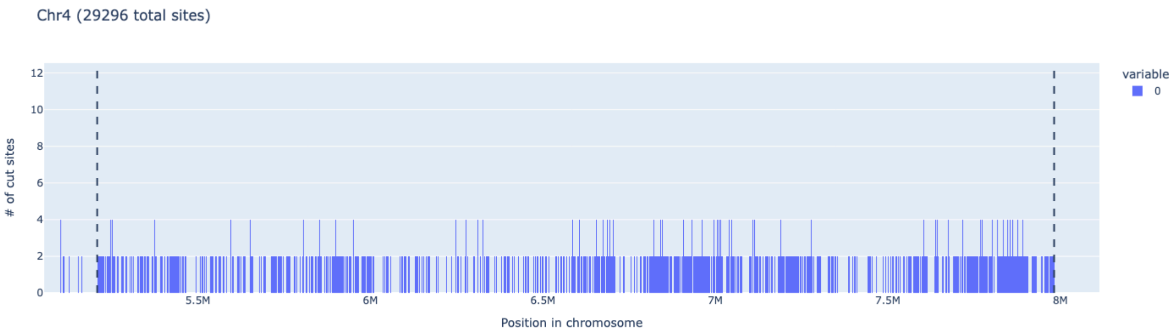
NdeI cutting profile in Chromosome 1 (centromere)



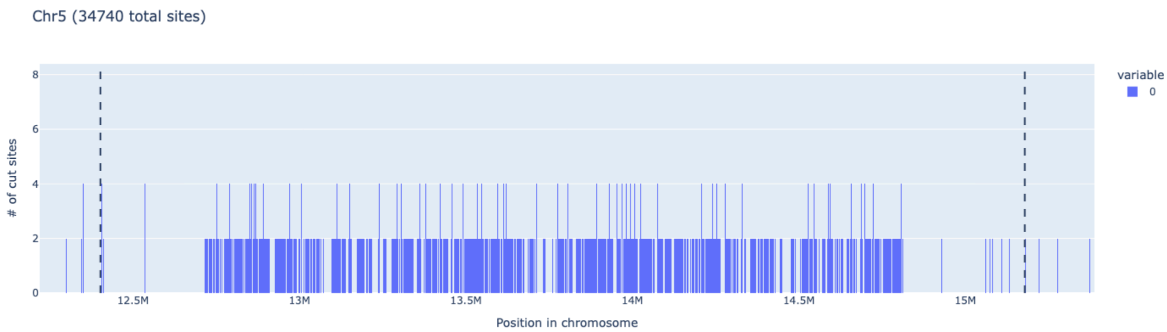
NdeI cutting profile in Chromosome 2 (centromere)



NdeI cutting profile in Chromosome 3 (centromere)



NdeI cutting profile in Chromosome 4 (centromere)



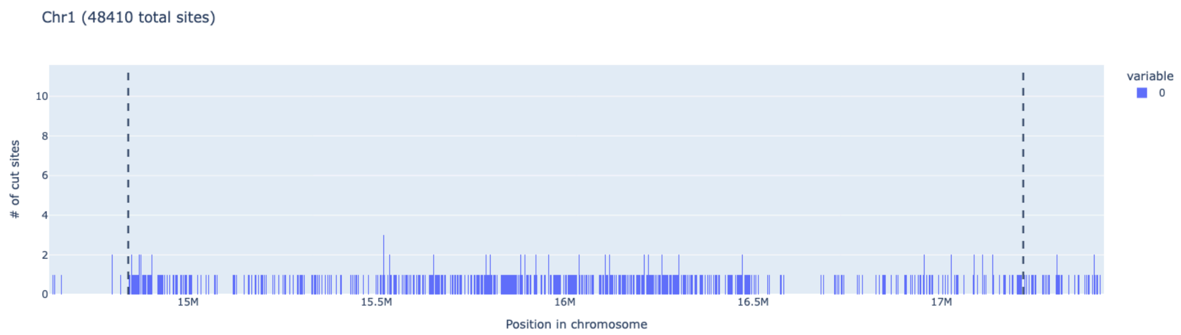
NdeI cutting profile in Chromosome 5 (centromere)

Chromosome	Enzyme	Enzyme Recogniti	Total Sites	Centromere Sites	Centromere Mean	Centromere	Arm Mean Spacing	Arm Spacing

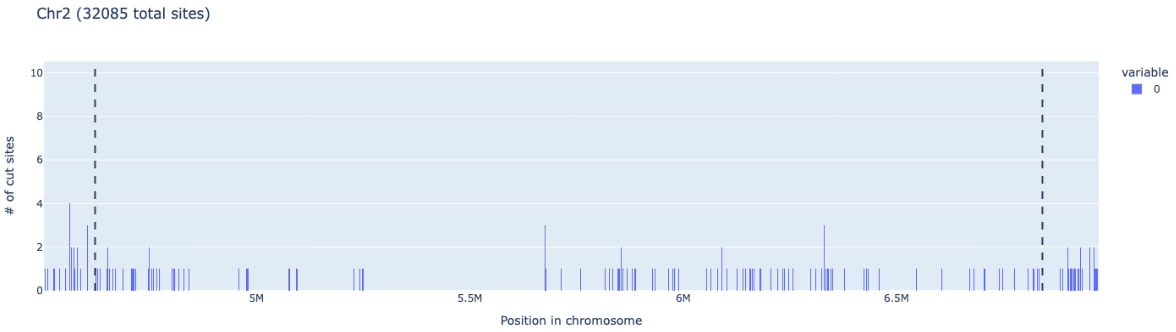
		on Sequence			Spacing (bp)	Spacing Std Dev	(bp)	Std Dev
Chr1	NdeI	CATATG	42328	22224	105.3061693	369.2805419	1623.092374	17038.03127
Chr2	NdeI	CATATG	31128	17394	127.615535	242.6344	1674.93694	19516.95032
Chr3	NdeI	CATATG	35222	19508	114.242272	251.5581187	1663.760771	18029.54584
Chr4	NdeI	CATATG	29296	16978	163.3857572	306.0408025	1832.455874	25555.39294
Chr5	NdeI	CATATG	34740	17186	161.2343323	638.3595872	1718.078904	21567.29202

- **AlwI**

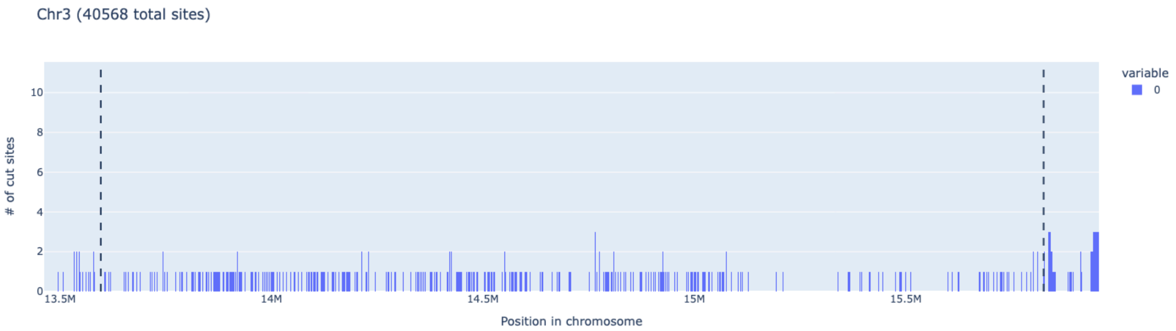
AlwI performance across all 5 chromosomes



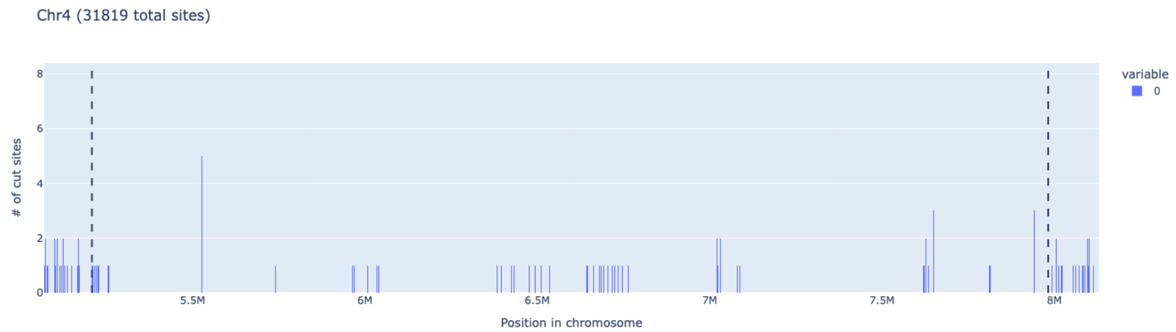
AlwI cutting profile in Chromosome 1 (centromere)



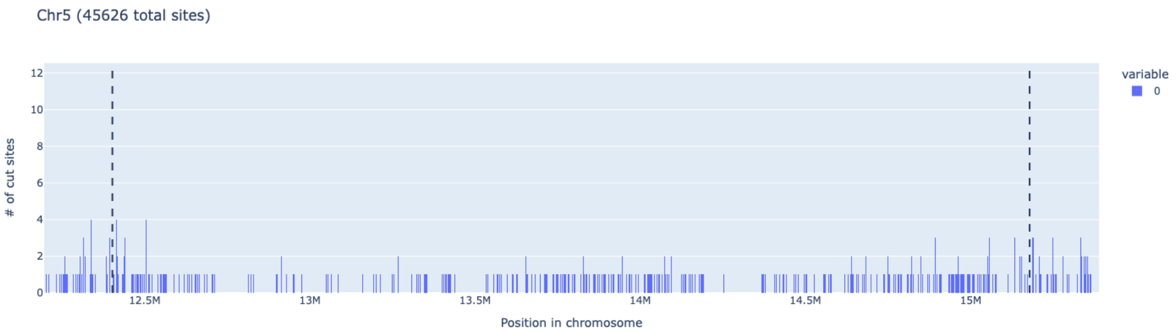
AlwI cutting profile in Chromosome 2 (centromere)



AlwI cutting profile in Chromosome 3 (centromere)



AlwI cutting profile in Chromosome 4 (centromere)

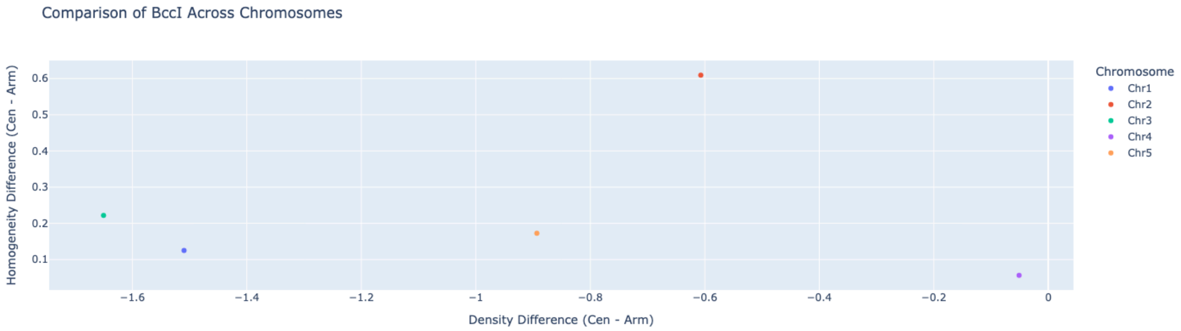


AlwI cutting profile in Chromosome 5 (centromere)

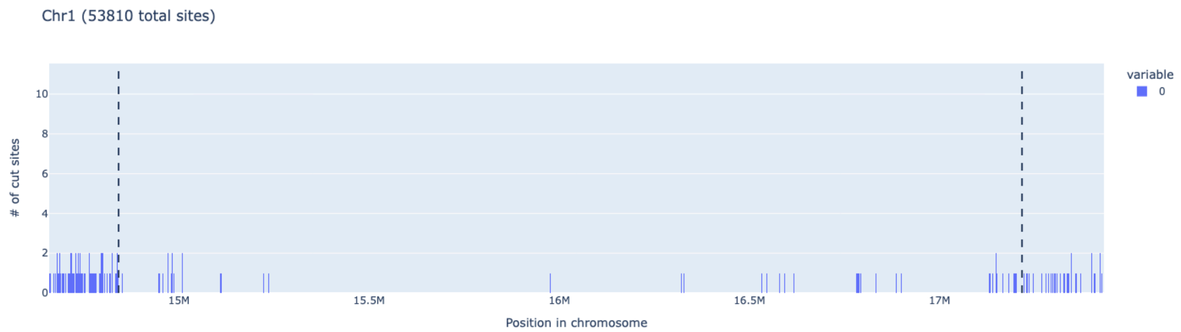
Chromos	Enzyme	Enzyme	Total	Centrom	Centrom	Centrom	Arm	Arm

ome		Recognition Sequence	Sites	ere Sites	ere Mean Spacing (bp)	ere Spacing Std Dev	Mean Spacing (bp)	Spacing Std Dev
Chr1	AlwI	GGATC	48410	5521	430.2512681	496.4657527	760.8022291	11520.39489
Chr2	AlwI	GGATC	32085	1161	1913.740517	5765.344294	743.9856741	12894.88431
Chr3	AlwI	GGATC	40568	3330	669.7122259	1309.954608	701.9969117	11587.55857
Chr4	AlwI	GGATC	31819	667	4141.013514	24703.92164	724.7710828	15770.55483
Chr5	AlwI	GGATC	45626	4143	670.266055	1302.31845	727.126344	13662.81288

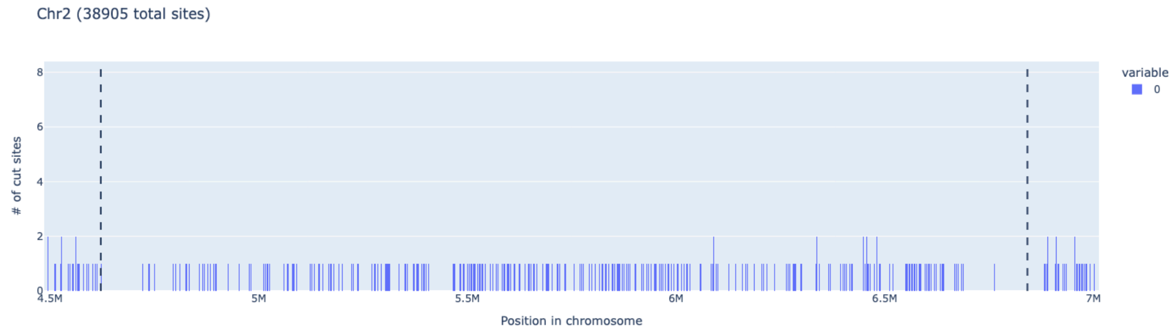
- **BccI**



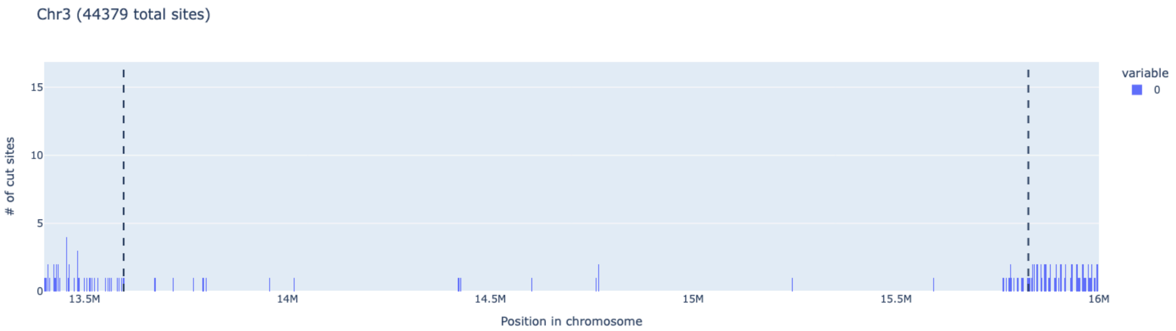
BccI performance across all 5 chromosomes



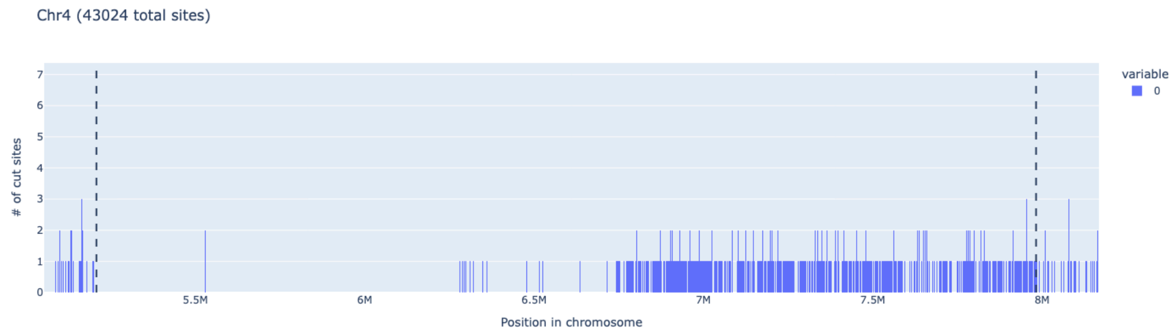
BccI cutting profile in Chromosome 1 (centromere)



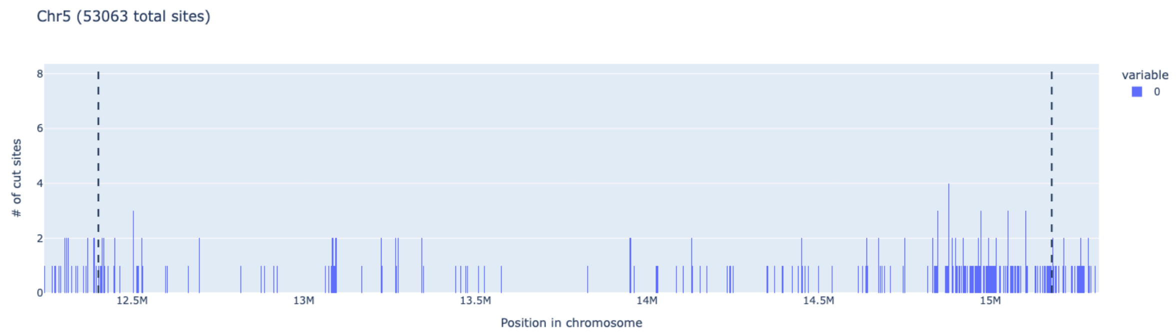
BccI cutting profile in Chromosome 2 (centromere)



BccI cutting profile in Chromosome 3 (centromere)



BccI cutting profile in Chromosome 4 (centromere)



BccI cutting profile in Chromosome 5 (centromere)

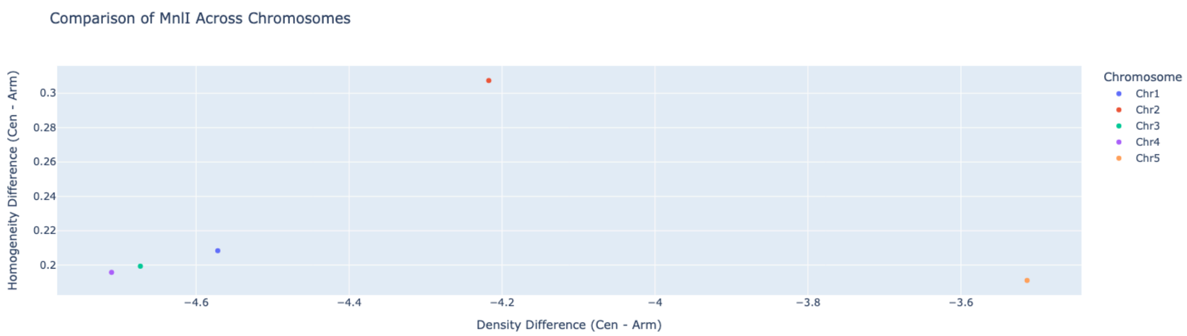
Chromosome	Enzyme	Enzyme Recogniti	Total Sites	Centromere Sites	Centromere Mean	Centromere	Arm Mean	Arm Spacing

		on Sequence			Spacing (bp)	Spacing Std Dev	Spacing (bp)	Std Dev
Chr1	BccI	CCATC	53810	591	4023.937288	21815.40116	613.1568642	10324.86545
Chr2	BccI	CCATC	38905	2536	875.8126233	1320.735552	632.570584	11771.90782
Chr3	BccI	CCATC	44379	418	5343.664269	19239.34681	594.6969745	10653.46908
Chr4	BccI	CCATC	43024	5161	531.0775194	5414.97547	596.2627965	14272.55898
Chr5	BccI	CCATC	53063	2631	1055.572243	4784.30516	598.0950209	12387.69915

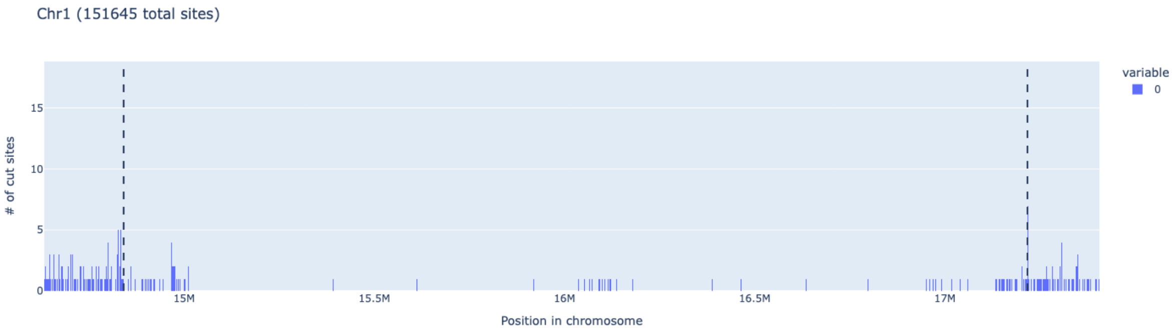
- **MluCI**

Chromosome	Enzyme	Enzyme Recognition Sequence	Total Sites	Centromere Sites	Centromere Mean Spacing (bp)	Centromere Spacing Std Dev	Arm Mean Spacing (bp)	Arm Spacing Std Dev
Chr1	MluCI	AATT	597106	18030	131.7663764	181.904739	56.35125847	3124.02877
Chr2	MluCI	AATT	394834	5750	385.9789529	1009.13376	59.13407165	3562.963392
Chr3	MluCI	AATT	455282	11900	187.3699471	332.8501862	58.96314005	3350.552129
Chr4	MluCI	AATT	363354	3094	895.748141	4005.396807	62.67311851	4623.781309
Chr5	MluCI	AATT	539450	15316	181.1467189	392.4720356	57.54906102	3836.888549

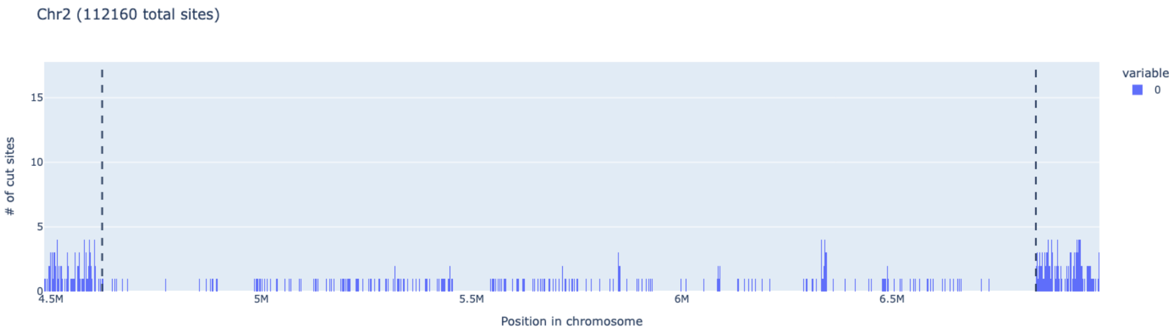
- **MnlI**



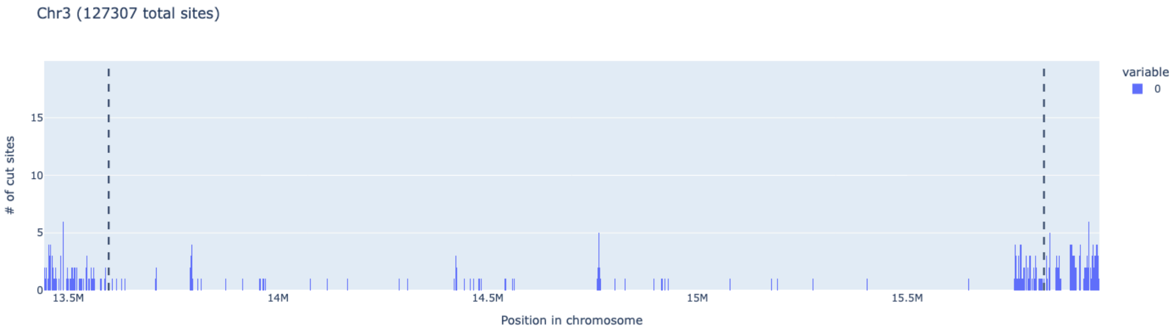
MnlI performance across all 5 chromosomes



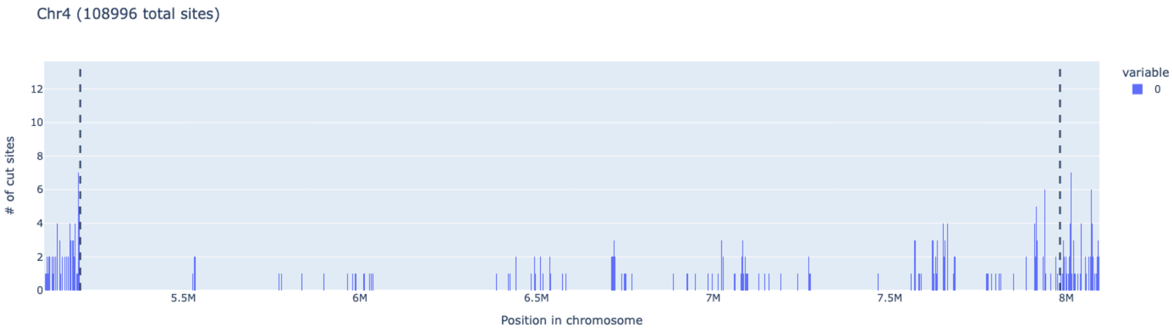
MnII cutting profile in Chromosome 1 (centromere)



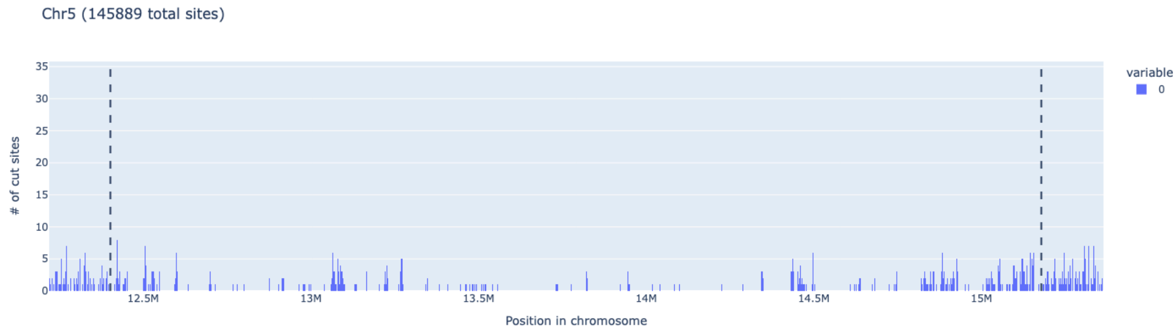
MnII cutting profile in Chromosome 2 (centromere)



MnII cutting profile in Chromosome 3 (centromere)



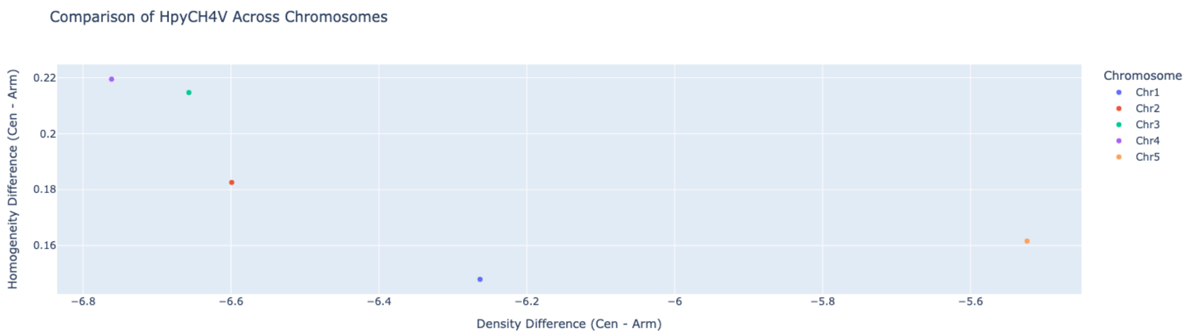
MnII cutting profile in Chromosome 4 (centromere)



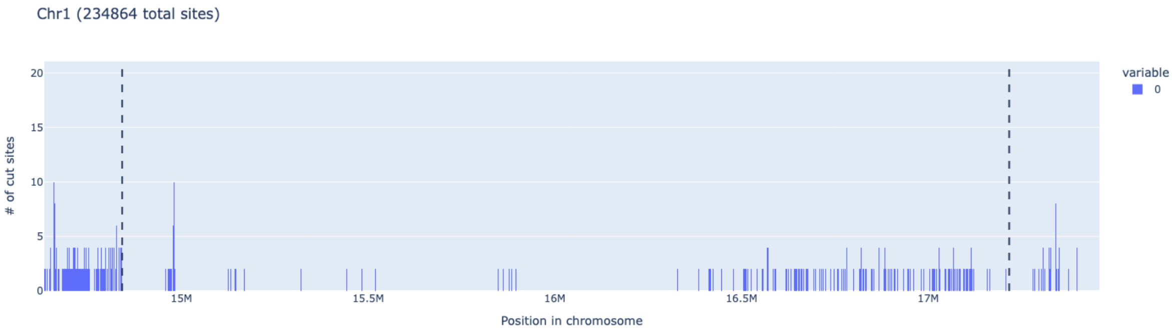
MnII cutting profile in Chromosome 5 (centromere)

Chromosome	**Enzyme**	**Enzyme Recognition Sequence**	**Total Sites**	**Centromere Sites**	**Centromere Mean Spacing (bp)**	**Centromere Spacing Std Dev**	**Arm Mean Spacing (bp)**	**Arm Spacing Std Dev**
Chr1	MnlI	CCTC	151645	967	2457.940994	10086.07175	216.5639945	6127.00157
Chr2	MnlI	CCTC	112160	2361	940.3927966	2776.627276	209.5545912	6705.018217
Chr3	MnlI	CCTC	127307	1324	1683.836735	7246.612811	207.5109222	6291.189172
Chr4	MnlI	CCTC	108996	1927	1439.263759	6524.391762	210.8817013	8482.330509
Chr5	MnlI	CCTC	145889	4568	607.3790234	2761.569541	213.439216	7394.47028

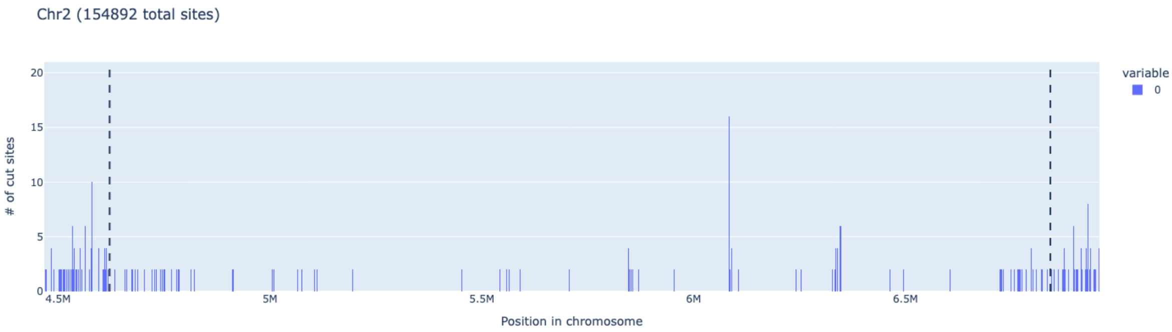
- ## HpyCH4V



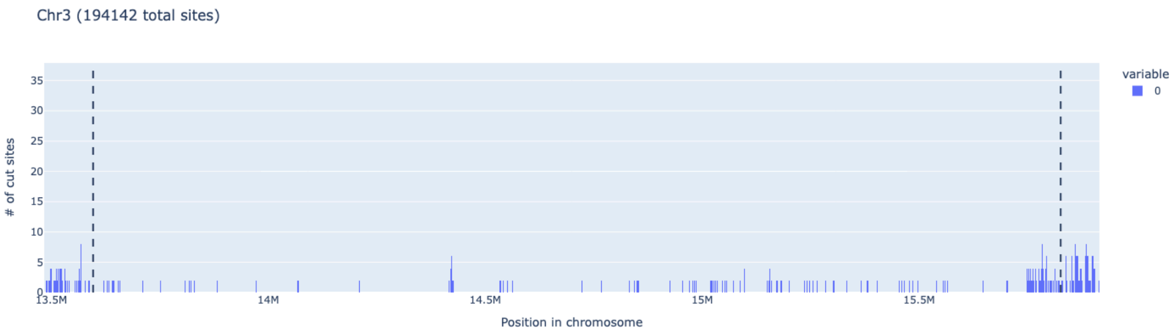
HpyCH4V performance across all 5 chromosomes



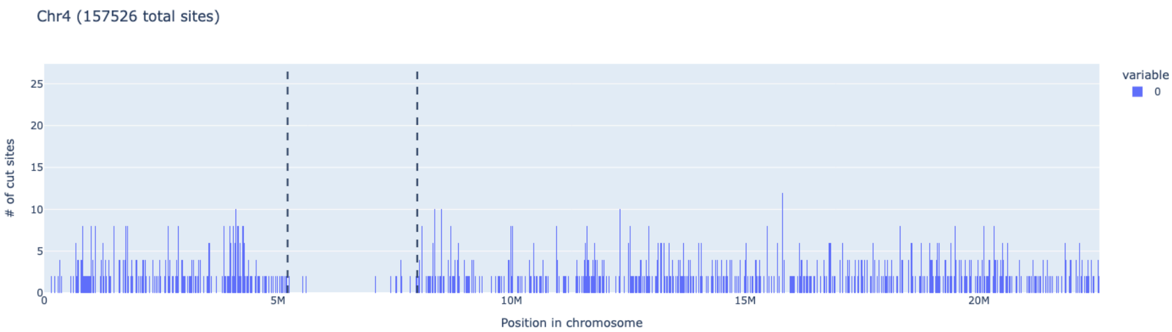
HpyCH4V cutting profile in Chromosome 1 (centromere)



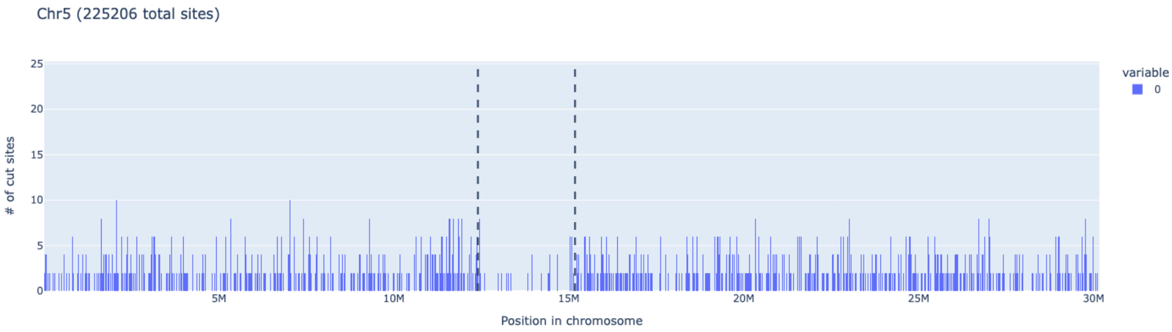
HpyCH4V cutting profile in Chromosome 2 (centromere)



HpyCH4V cutting profile in Chromosome 3 (centromere)



HpyCH4V cutting profile in Chromosome 4 (centromere)



HpyCH4V cutting profile in Chromosome 5 (centromere)

Chromosome	Enzyme	Enzyme Recognition Sequence	Total Sites	Centromere Sites	Centromere Mean Spacing (bp)	Centromere Spacing Std Dev	Arm Mean Spacing (bp)	Arm Spacing Std Dev
Chr1	HpyCH4V	TGCA	234864	3298	718.9035487	4075.082156	140.9178373	4945.54055
Chr2	HpyCH4V	TGCA	154892	1706	1299.233431	6216.508622	150.183719	5680.87991
Chr3	HpyCH4V	TGCA	194142	2976	748.3751261	3098.748247	136.7473439	5105.441413
Chr4	HpyCH4V	TGCA	157526	2898	956.2105627	3980.882974	146.0186255	7059.587846
Chr5	HpyCH4V	TGCA	225206	6800	407.6037653	2205.794841	138.1076715	5947.075763